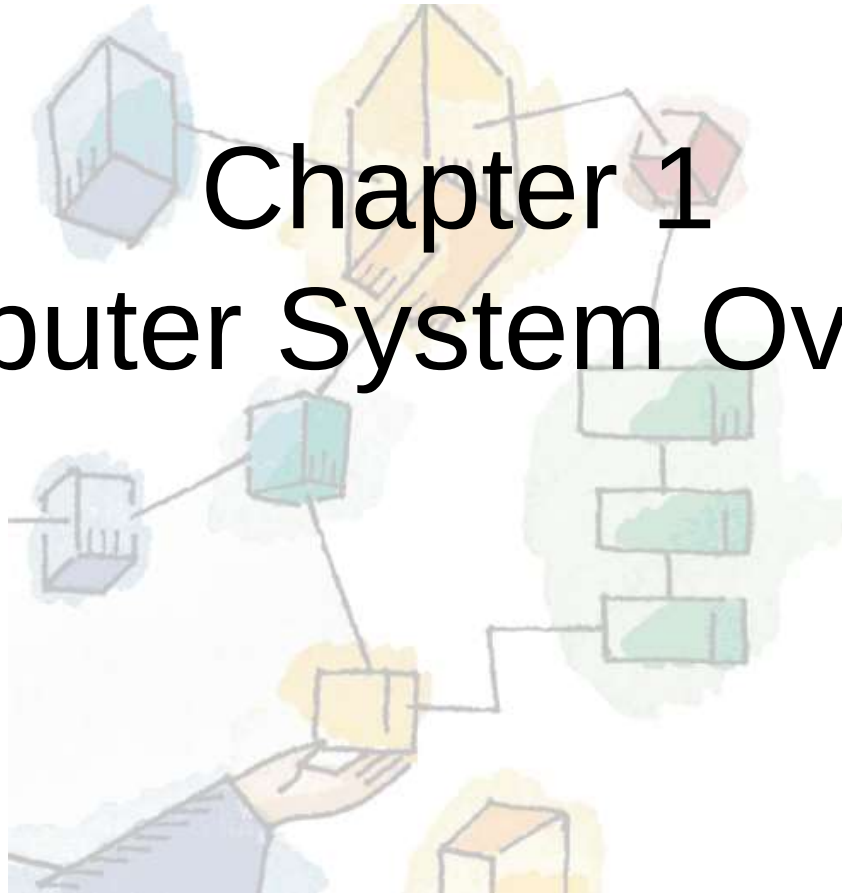


*Operating Systems:
Internals and Design Principles, 6/E*
William Stallings

Chapter 1

Computer System Overview



Patricia Roy
Manatee Community College, Venice,
FL

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*Operativni sistemi:
Principi unutrašnje organizacije i
dizaja, 6/E*
William Stallings

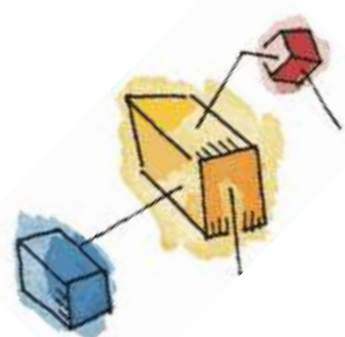
Poglavlje 1

Pregled računarskog sistema



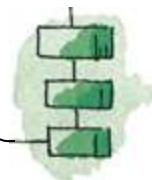
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FL

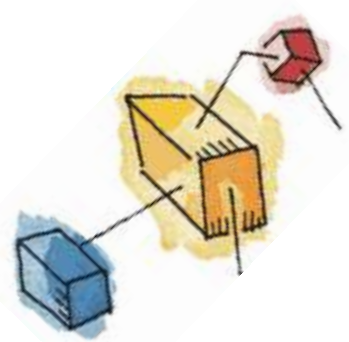
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Operativni sistem

- Koristi resurse hardvera jednog ili više procesora
- Obezbeđuje skup servisa korisnicima sistema
- Upravlja sekundarnom memorijom i U/I uređajima

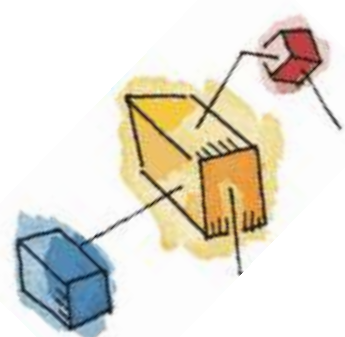




Operating System

- Exploits the hardware resources of one or more processors
- Provides a set of services to system users
- Manages secondary memory and I/O devices

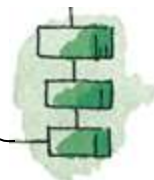


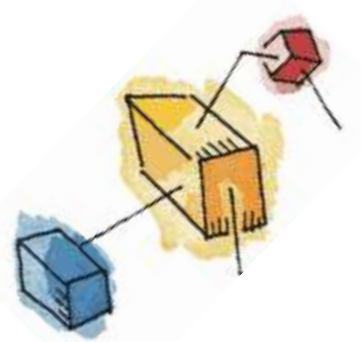


1.1 Osnovni elementi

- Procesor
- CPU (*Central Processing Unit*) –kada postoji samo jedan procesor)
 - U/I adresni registar
 - U/I bafer registar

– Str.9

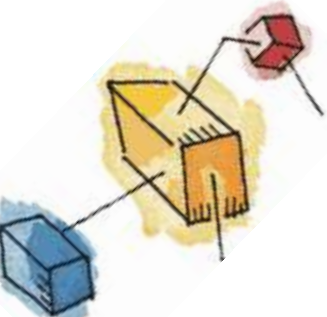




Osnovni elementi

- Glavna (osnovna) memorija
 - Izbrisiva (nestalna)
 - Naziva se i stvarna memorija ili osnovna (*primary*) memory

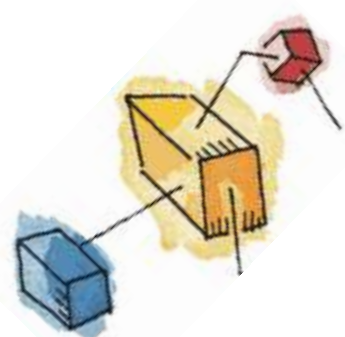




Basic Elements

- Main Memory
 - Volatile
 - Referred to as real memory or primary memory

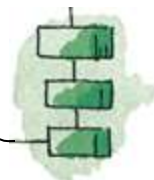


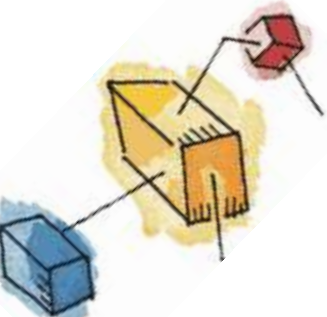


Osnovni elementi

- U/I moduli
 - Secundarna memorija
 - Komunikaciona oprema
 - Terminali
- Sistemska magistrala
 - Omogućava komunikaciju između procesora, glavne memorije, i U/I modula

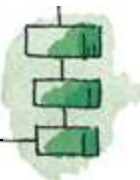
– Str.9





Basic Elements

- I/O Modules
 - Secondary Memory Devices
 - Communications equipment
 - Terminals
- System bus
 - Communication among processors, main memory, and I/O modules



Komponente računara: pogled sa najvišeg nivoa

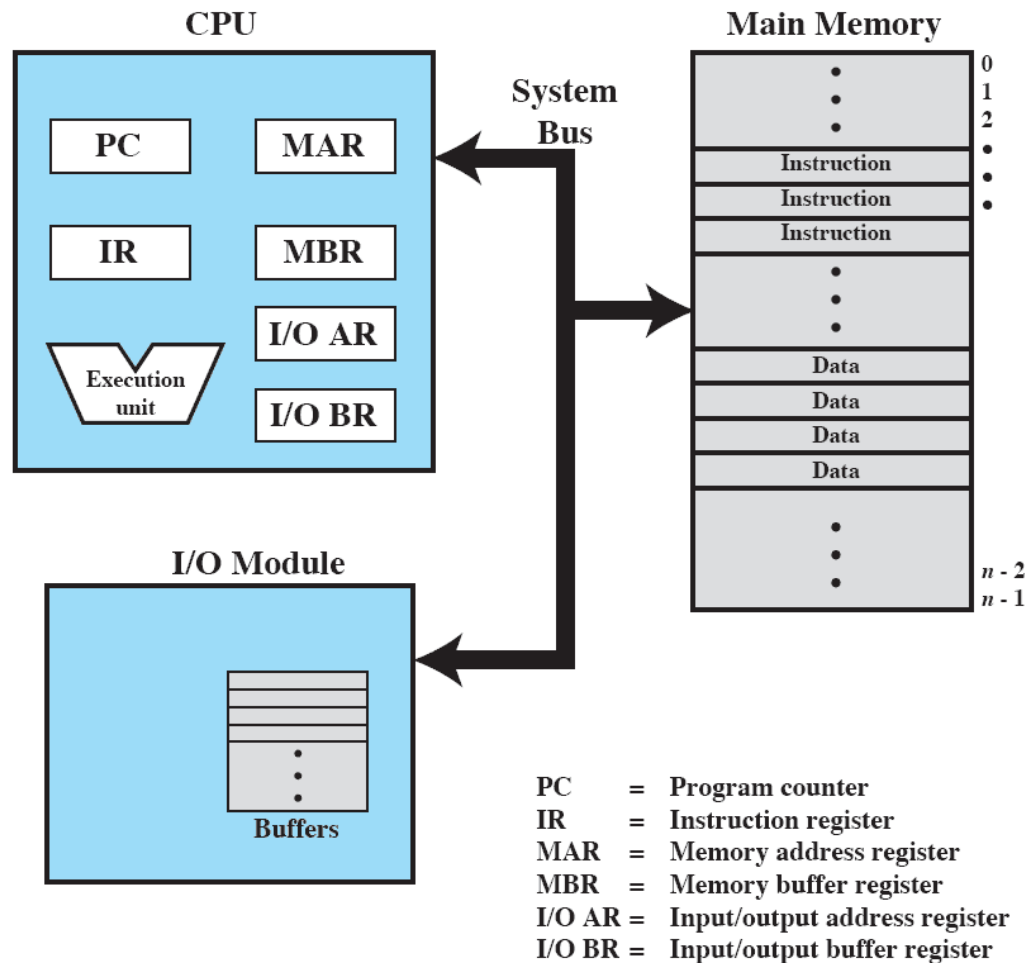


Figure 1.1 Computer Components: Top-Level View

Computer Components: Top-Level View

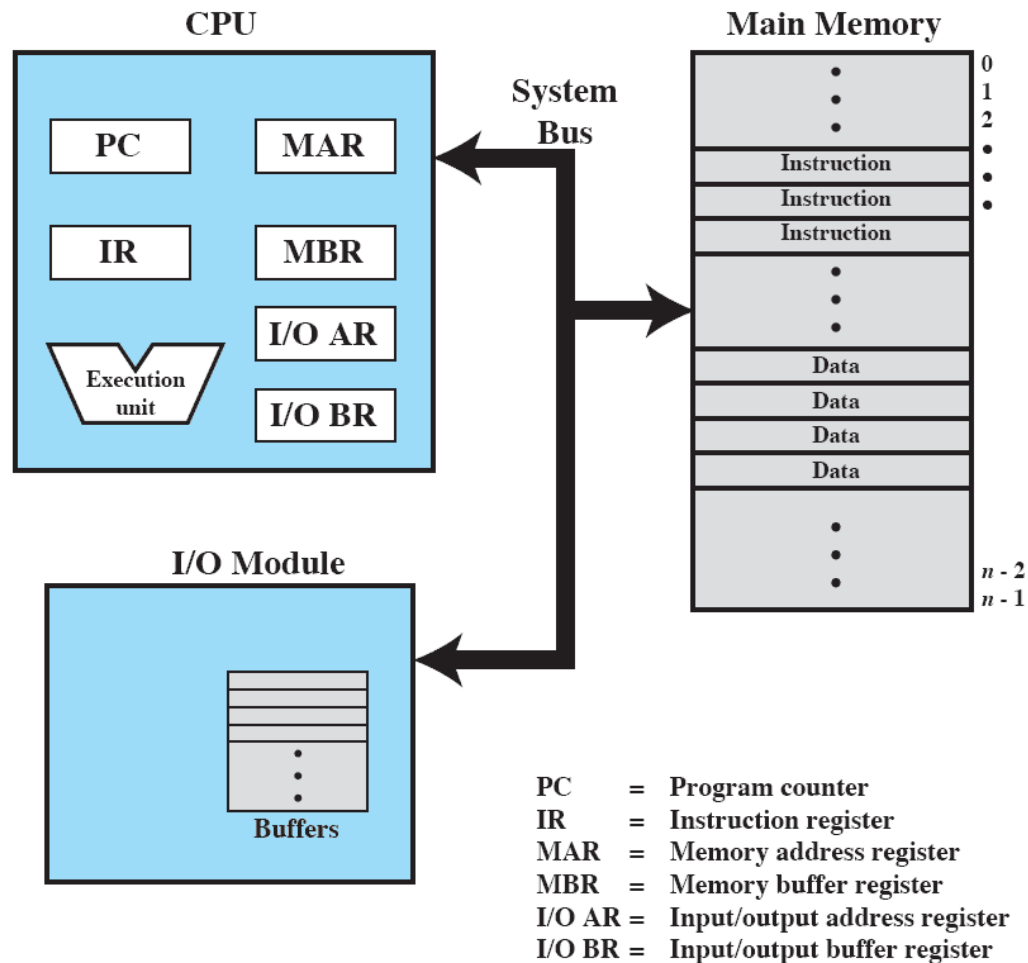
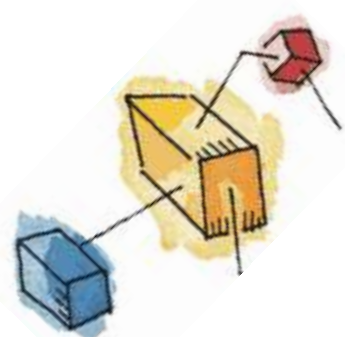
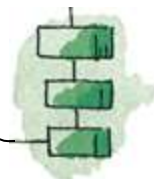


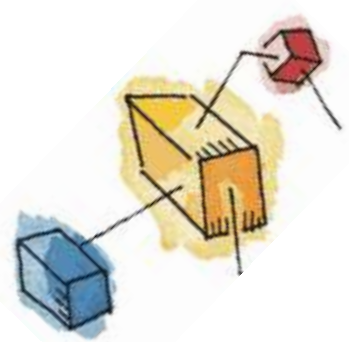
Figure 1.1 Computer Components: Top-Level View



1.2 Procesorski registri

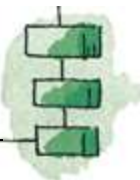
- Registri vidljivi korisniku
 - Omogućavaju programeru da minimizuje adresiranje glavne memorije optimizacijom upotrebe registara
- Upravljački i statusni registri
 - Koriste ih procesor da bi upravljao operacijama procesora
 - Koriste ih privilegovane OS rutine da bi upravljale izvršavanjem programa

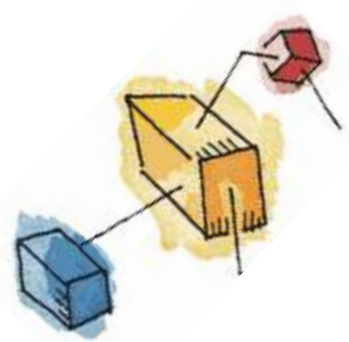




1.2 Procesorski registri

- User-visible registers
 - Enable programmer to minimize main memory references by optimizing register use
- Control and status registers
 - Used by processor to control operating of the processor
 - Used by privileged OS routines to control the execution of programs

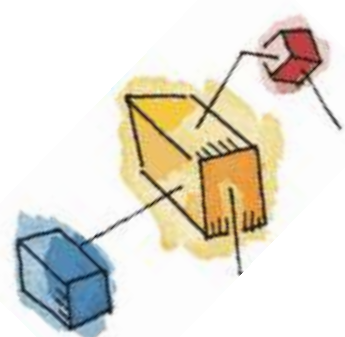




User-Visible Registers

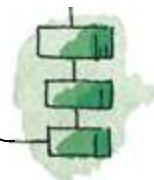
- May be referenced by machine language
- Available to all programs – application programs and system programs

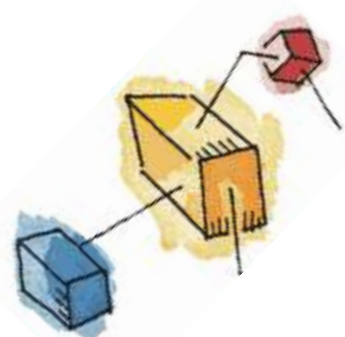




Registri vidljivi korisniku

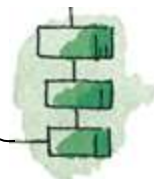
- Mogu se adresirati pomoću mašinskog jezika
- Dostupni su svim programima, kako aplikativnim tako i sistemskim programima

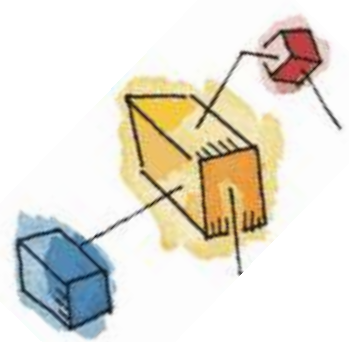




Registri vidljivi korisniku

- Registri podataka
- Adreni registri
 - Indeksni registar: Dodaje indeks osnovnoj vrednosti kako bi se dobila efektivna adresa
 - Pokazivač (*pointer*) segmenta: Kada je memoija podeljena u delove, memorijska referenca sastoji se od adrese određenog segmenta i pomeraja (*offset*) unutar njega
 - Pokazivač steka: Pokazuje na vrh steka

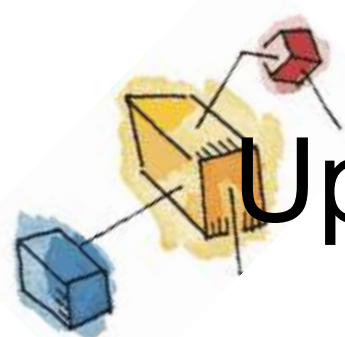




User-Visible Registers

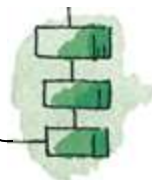
- Data
- Address
 - Index register: Adding an index to a base value to get the effective address
 - Segment pointer: When memory is divided into segments, memory is referenced by a segment and an offset
 - Stack pointer: Points to top of stack

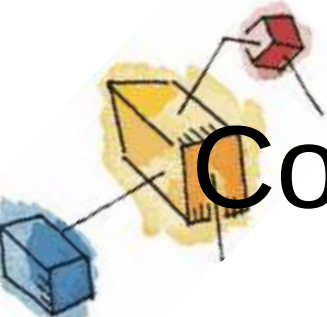




Upravljački i statusni registri

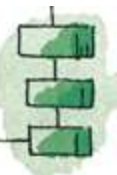
- Programski brojač (*Program counter (PC)*)
 - Sadrži adresu sledeće instrukcije za obradu
- Instrukcijski registar (IR)
 - Sadrži kod instrukcije koja se izvršava
- Statusni registar (PSW) (*Program status word*)
 - Sadrži statusne informacije

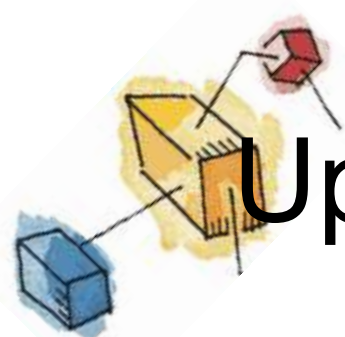




Control and Status Registers

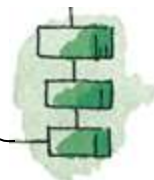
- Program counter (PC)
 - Contains the address of an instruction to be fetched
- Instruction register (IR)
 - Contains the instruction most recently fetched
- Program status word (PSW)
 - Contains status information
 - Sadrži uslovne kodove
 - Bit zabrane prekida
 - Bit režima supervizor/korisnik

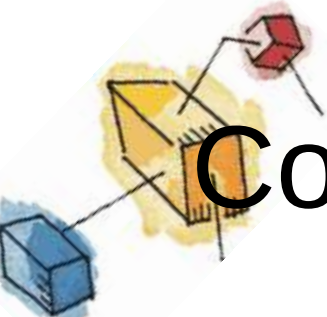




Upravljački i statusni registri

- Uslovni kodovi ili markeri (*flags*)
 - Bitovi koje obično postavlja hardver procesora kao rezultat operacija
 - Primer (aritmetička operacija može da proizvede)
 - Pozitivan, negativan, nulu ili prekoračenje (*overflow*) rezultat

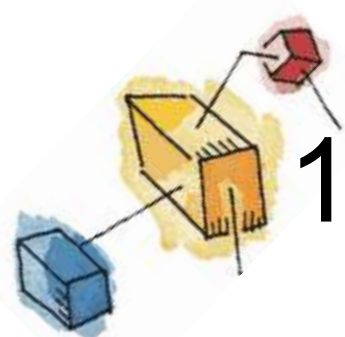




Control and Status Registers

- Condition codes or flags
 - Bits set by processor hardware as a result of operations
 - Example
 - Positive, negative, zero, or overflow result

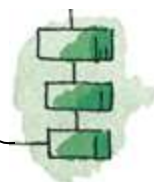


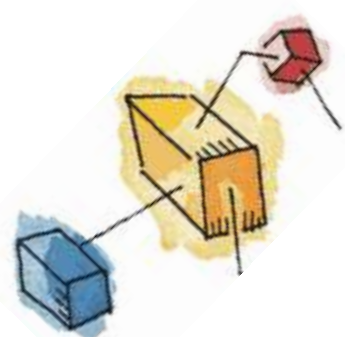


1.3 Izvršavanje instrukcija

- Dva koraka
 - Procesor čita (*fetches*) instrukcije iz memorije
 - Procesor izvršava (*executes*) svaku naredbu

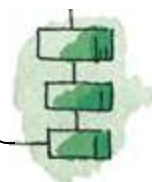
– Str.13

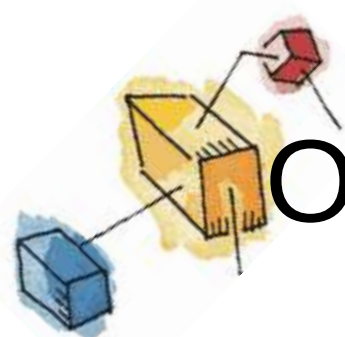




Instruction Execution

- Two steps
 - Processor reads (fetches) instructions from memory
 - Processor executes each instruction





Osnovni instrukcijski ciklus

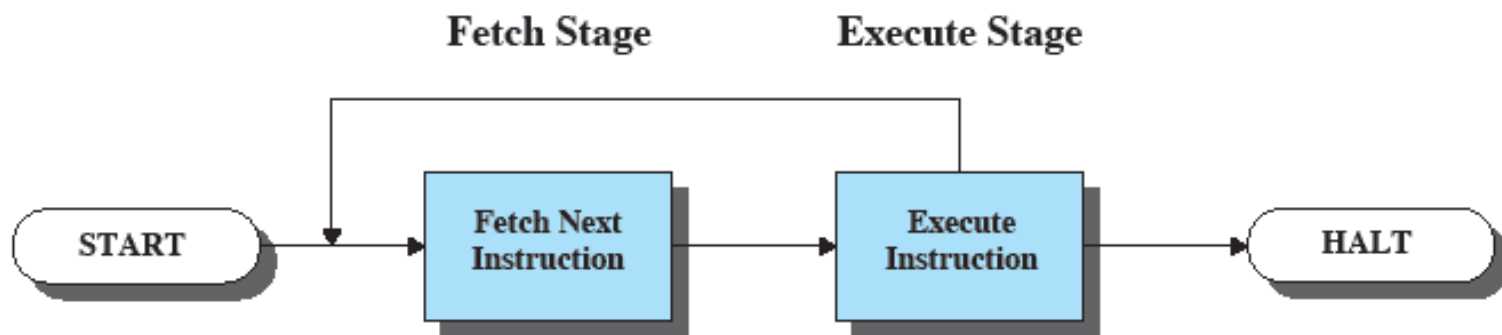
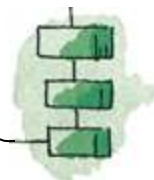


Figure 1.2 Basic Instruction Cycle



Basic Instruction Cycle

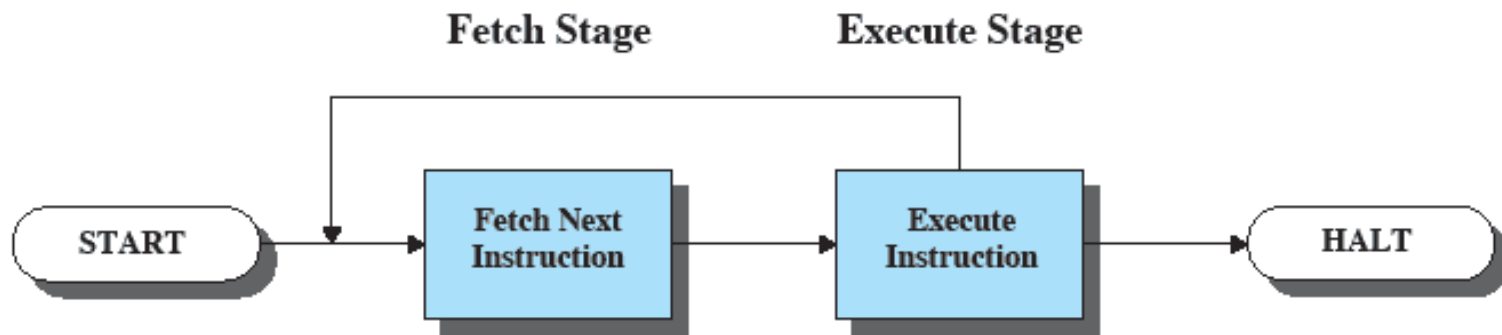


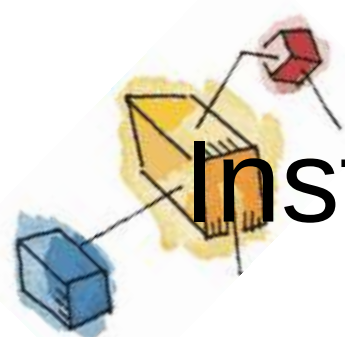
Figure 1.2 Basic Instruction Cycle



Pribavljanje i izvršavanje naredbi

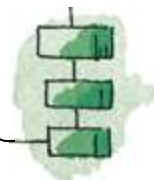
- Procesor pribavlja (čita, donosi) (*fetches*) naredbe iz memorije
- Programski brojač (*PC- program counter*) sadrži adresu sledeće naredbe za donošenje
- PC se uvećava (inkrementira) posle svakog donošenja naredbe

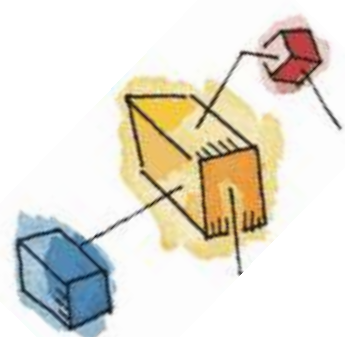




Instruction Fetch and Execute

- The processor fetches the instruction from memory
- Program counter (PC) holds address of the instruction to be fetched next
- PC is incremented after each fetch

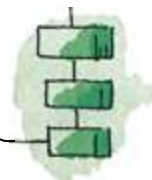


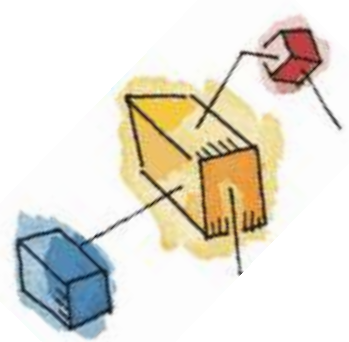


Registar naredbi - RN

Instruction Register

- Donešene naredbe se učitavaju u registar naredbi (*instruction register*)
- Kategorije radnji
 1. Procesor-memorija,
 2. procesor-U/I,
 3. Obrada podataka,
 4. Upravljanje





Instruction Register

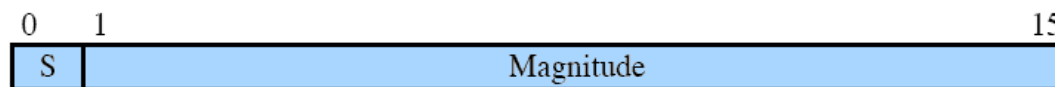
- Fetched instruction loaded into instruction register
- Categories
 - Processor-memory, processor-I/O, data processing, control



Karakteristike hipotetičkog računara



(a) Instruction format



(b) Integer format

Program counter (PC) = Address of instruction
Instruction register (IR) = Instruction being executed
Accumulator (AC) = Temporary storage

(c) Internal CPU registers

0001 = Load AC from memory
0010 = Store AC to memory
0101 = Add to AC from memory

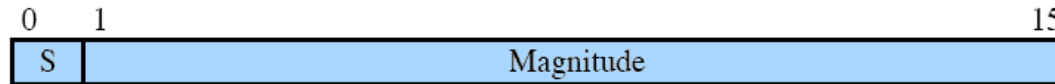
(d) Partial list of opcodes

Figure 1.3 Characteristics of a Hypothetical Machine

Characteristics of a Hypothetical Machine



(a) Instruction format



(b) Integer format

Program counter (PC) = Address of instruction
Instruction register (IR) = Instruction being executed
Accumulator (AC) = Temporary storage

(c) Internal CPU registers

0001 = Load AC from memory
0010 = Store AC to memory
0101 = Add to AC from memory

(d) Partial list of opcodes

Figure 1.3 Characteristics of a Hypothetical Machine

Primer izvršavanja programa

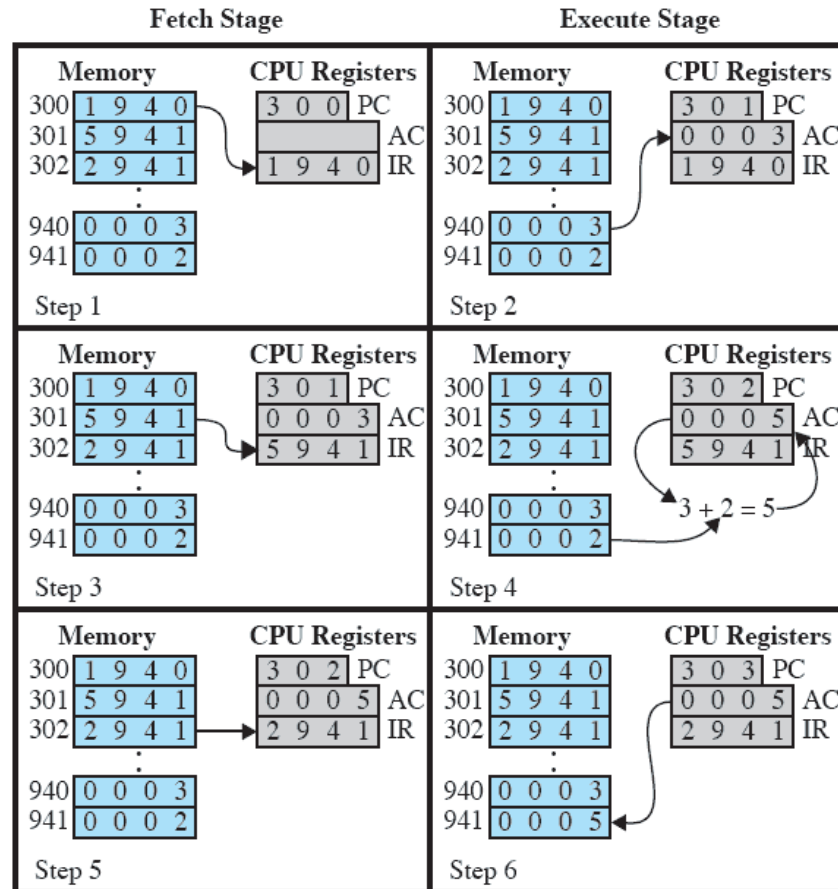


Figure 1.4 Example of Program Execution
(contents of memory and registers in hexadecimal)

Example of Program Execution

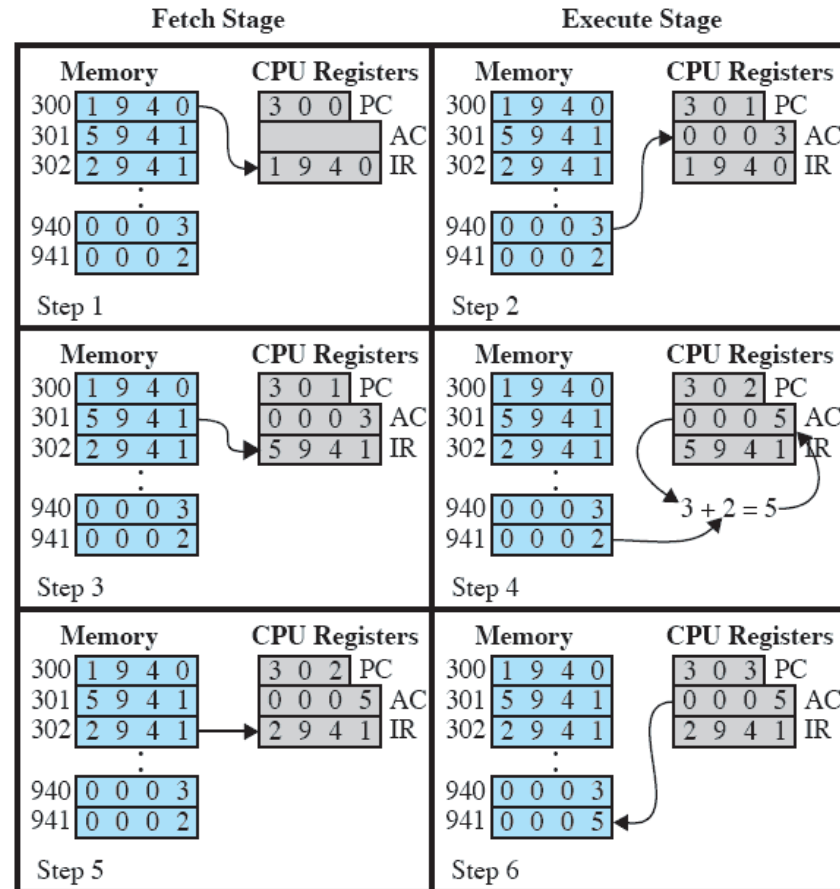
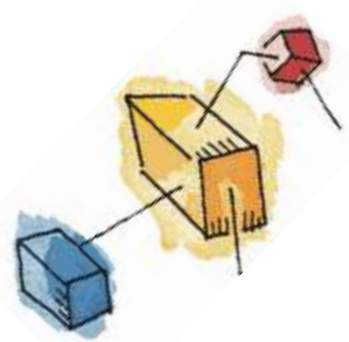


Figure 1.4 Example of Program Execution
(contents of memory and registers in hexadecimal)



1.4 Prekidi (*Interrupts*)

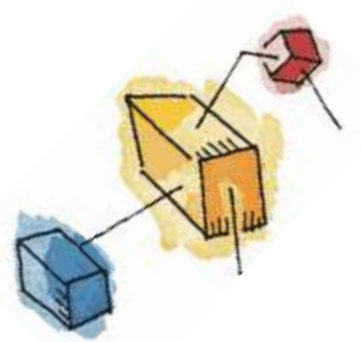
- Prekida normalan redosled rada procesora
- Većina U/I uređaja je mnogo sporija od procesora
 - Procesor mora da pravi pauzu dok čeka na U/I uređaj
 - To je veoma neproduktivna upotreba procesora

• Str.16

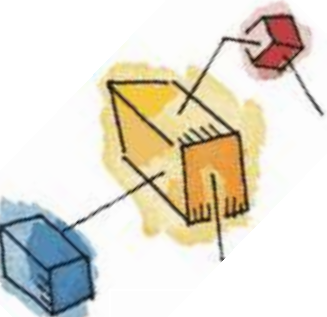


Interrupts

- Interrupt the normal sequencing of the processor
- Most I/O devices are slower than the processor
 - Processor must pause to wait for device



Klase prekida



Program

Generiše se zbog nekih uslova koji se pojavljuju kao rezultat izvršavanja naredbi, kao što je aritmetičko prekoračenje, deljenje nulom, pokušaj izvršavanja nepostojeće mašinske naredbe i adresiranje memorijskog prostora koji je dodeljen drugom korisniku.

Tajmer

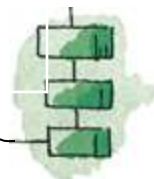
Generiše ga tajmer unutar rocesora. Ovim se operativnom sistemu dozvoljava redovno izvršavanje određenih funkcija.

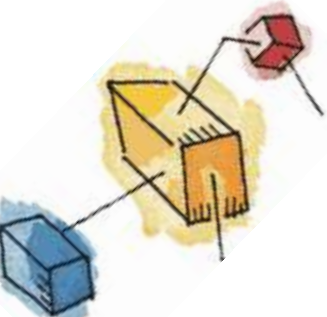
U/I

Generiše ga U/I kontroler kako bi signalizirao završetak operacije ili razne greške.

Otlaz hardvera

Generiše ga otkaz, kao što je nestanak struje ili greška pariteta memorije.

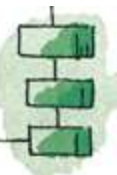




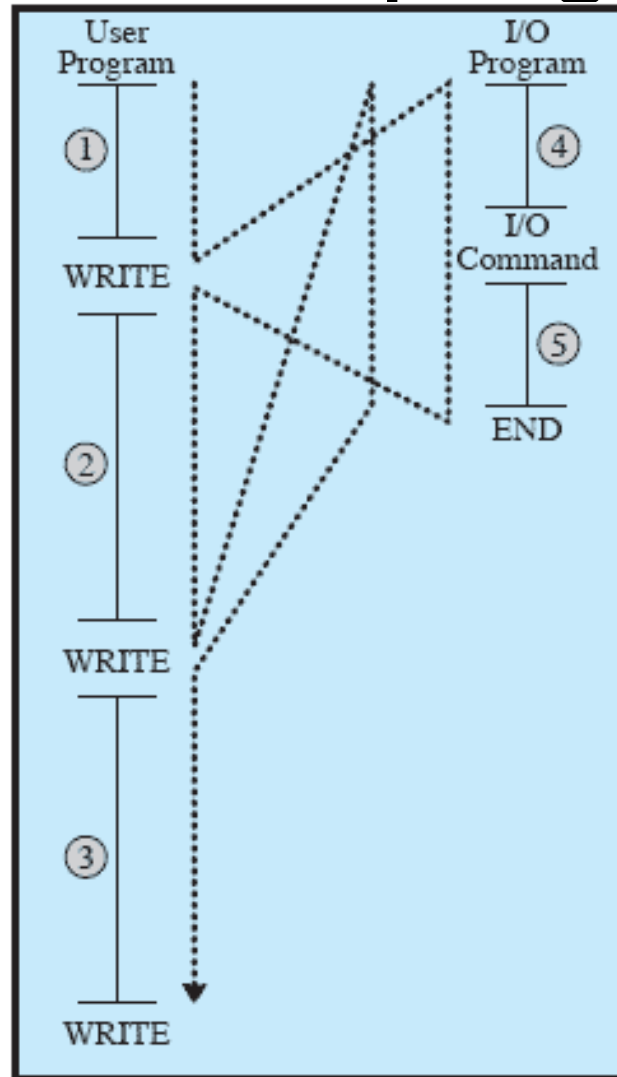
Classes of Interrupts

Table 1.1 **Classes of Interrupts**

Program	Generated by some condition that occurs as a result of an instruction execution, such as arithmetic overflow, division by zero, attempt to execute an illegal machine instruction, and reference outside a user's allowed memory space.
Timer	Generated by a timer within the processor. This allows the operating system to perform certain functions on a regular basis.
I/O	Generated by an I/O controller, to signal normal completion of an operation or to signal a variety of error conditions.
Hardware failure	Generated by a failure, such as power failure or memory parity error.

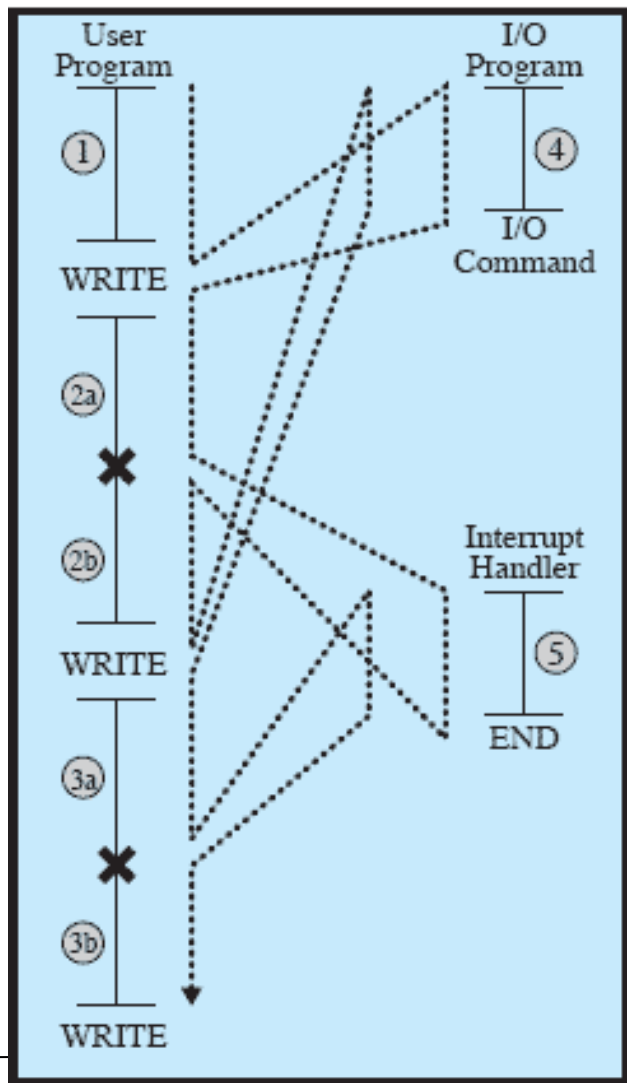


Tok kontrole programa



(a) No interrupts

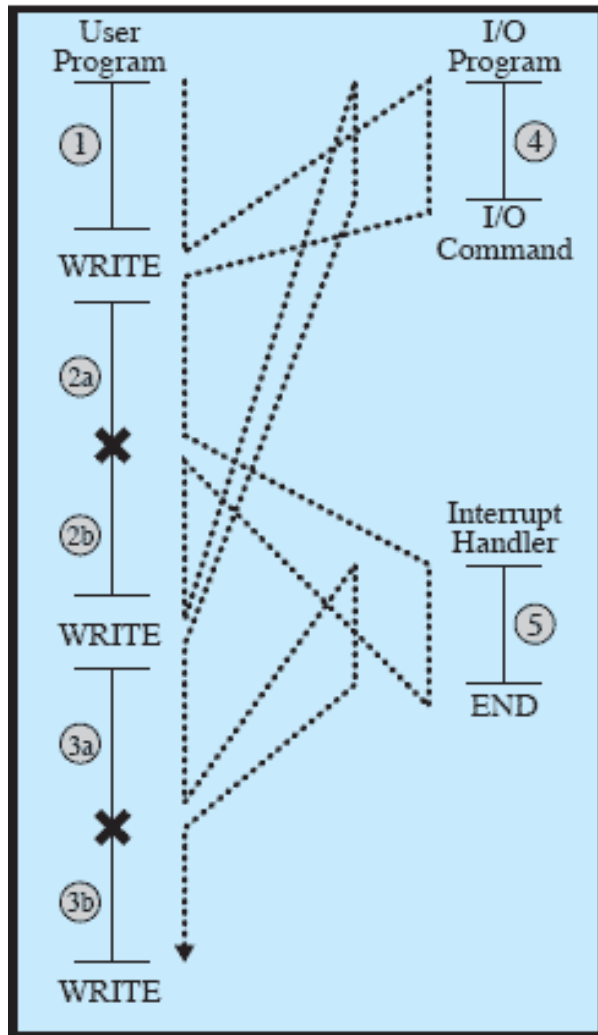
Tok kontrole programa



(b) Interrupts; short I/O wait

zahtev za prekidom
opsluživatelj prekida

Tok kontrole programa

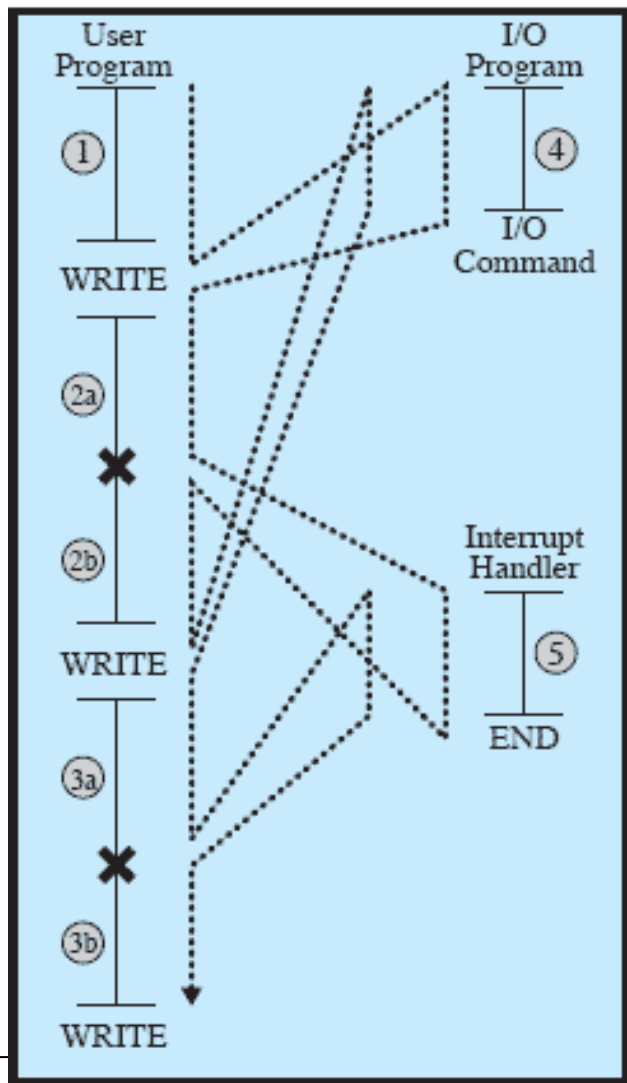


(b) Interrupts; short I/O wait

Prekidi i instrukcijski ciklusi

- zahtev za prekidom
- opsluživač prekida
 - ISTOVREMENO izvršavanje
 1. U/I operacije i
 2. (instrukcija) korisničkog programa
- Str.17.

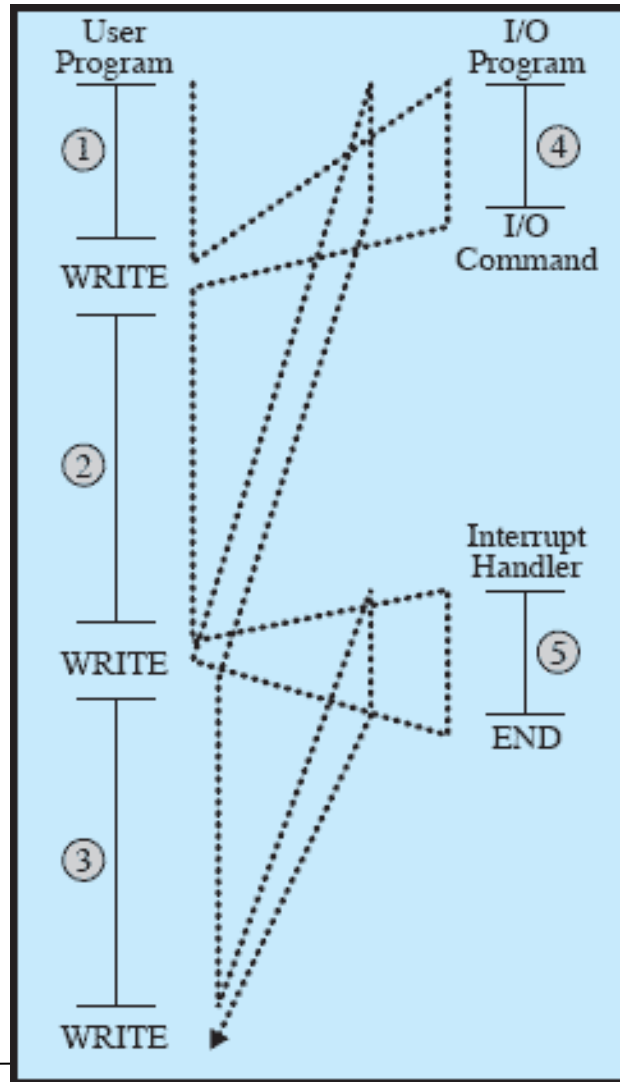
Tok kontrole programa



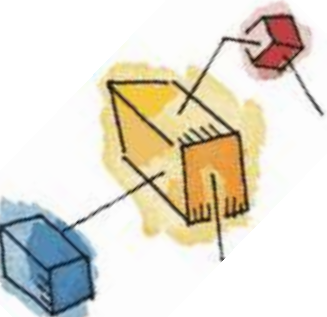
(b) Interrupts; short I/O wait

zahtev za prekidom
opsluživatelj prekida

Tok kontrole programa



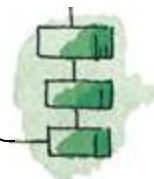
(c) Interrupts; long I/O wait

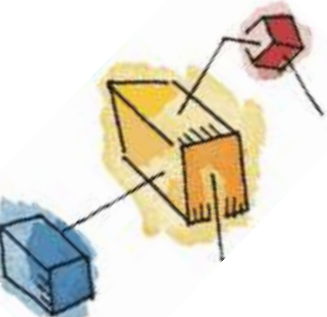


Prekidno stanje

- Procesor proverava pojavu interapta
- Ukoliko prekid čeka obradu procesor:
 - Prekida izvršavanje tekućeg programa
 - Izvršava rutinu za izvršavanje prekida

– Str.18





Interrupt Stage

- Processor checks for interrupts
- If interrupt
 - Suspend execution of program
 - Execute interrupt-handler routine



Prebacivanje kontrole pomoću prekida

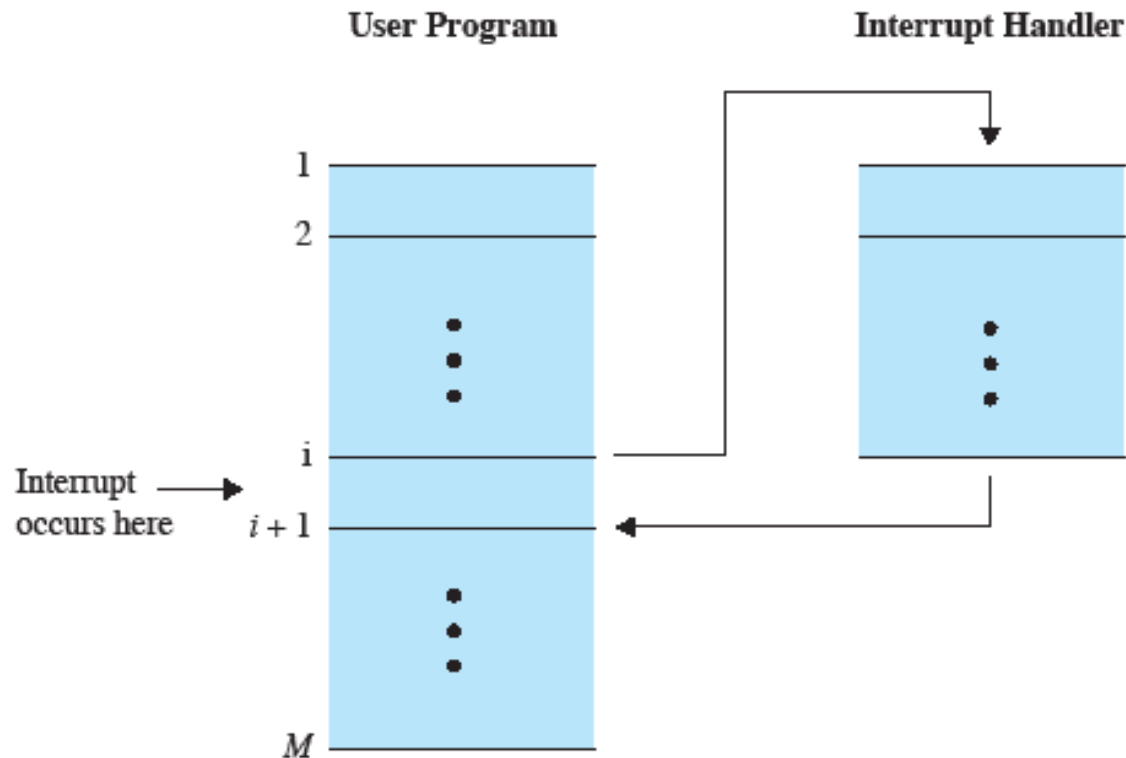


Figure 1.6 Transfer of Control via Interrupts

Transfer of Control via Interrupts

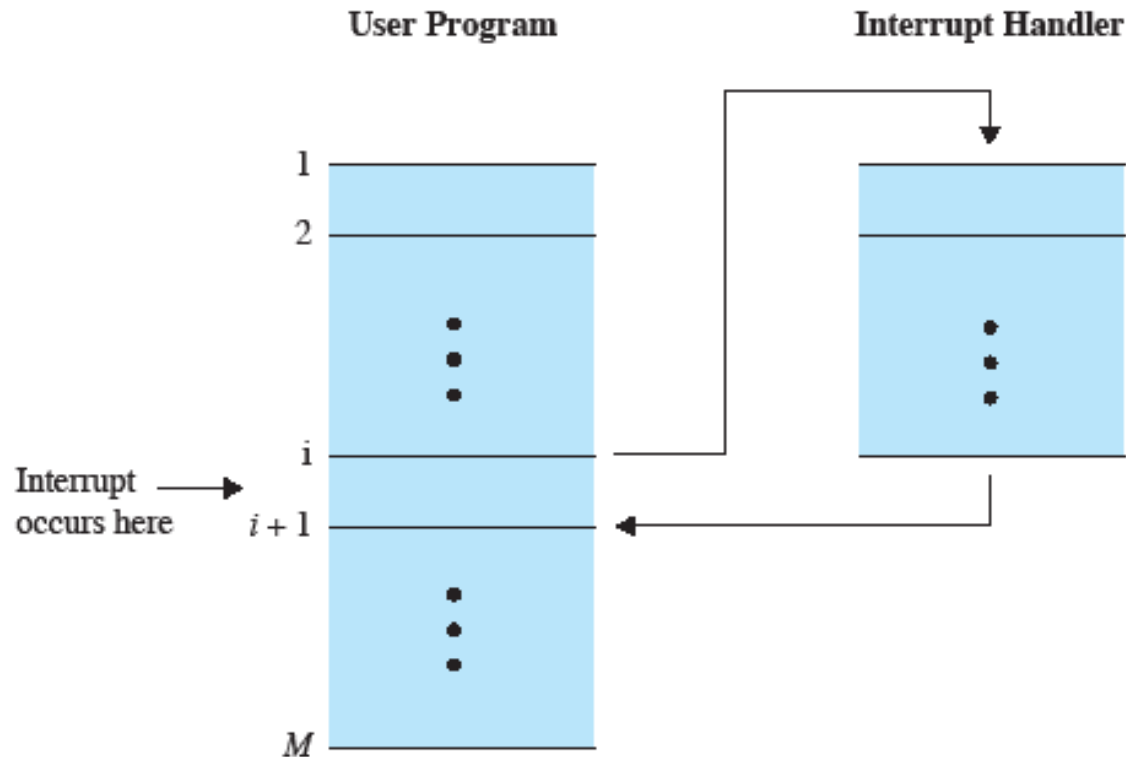


Figure 1.6 Transfer of Control via Interrupts

Instrukcijski ciklus sa prekidima

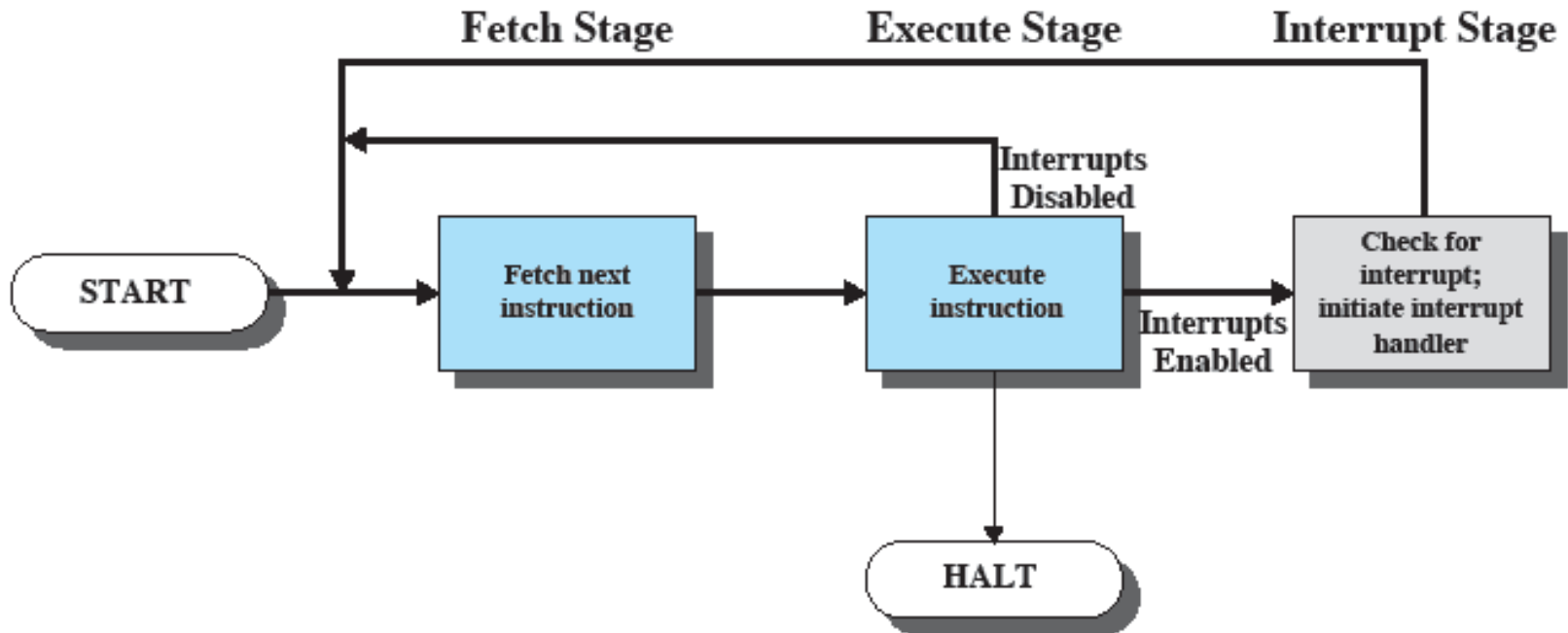


Figure 1.7 Instruction Cycle with Interrupts

Instruction Cycle with Interrupts

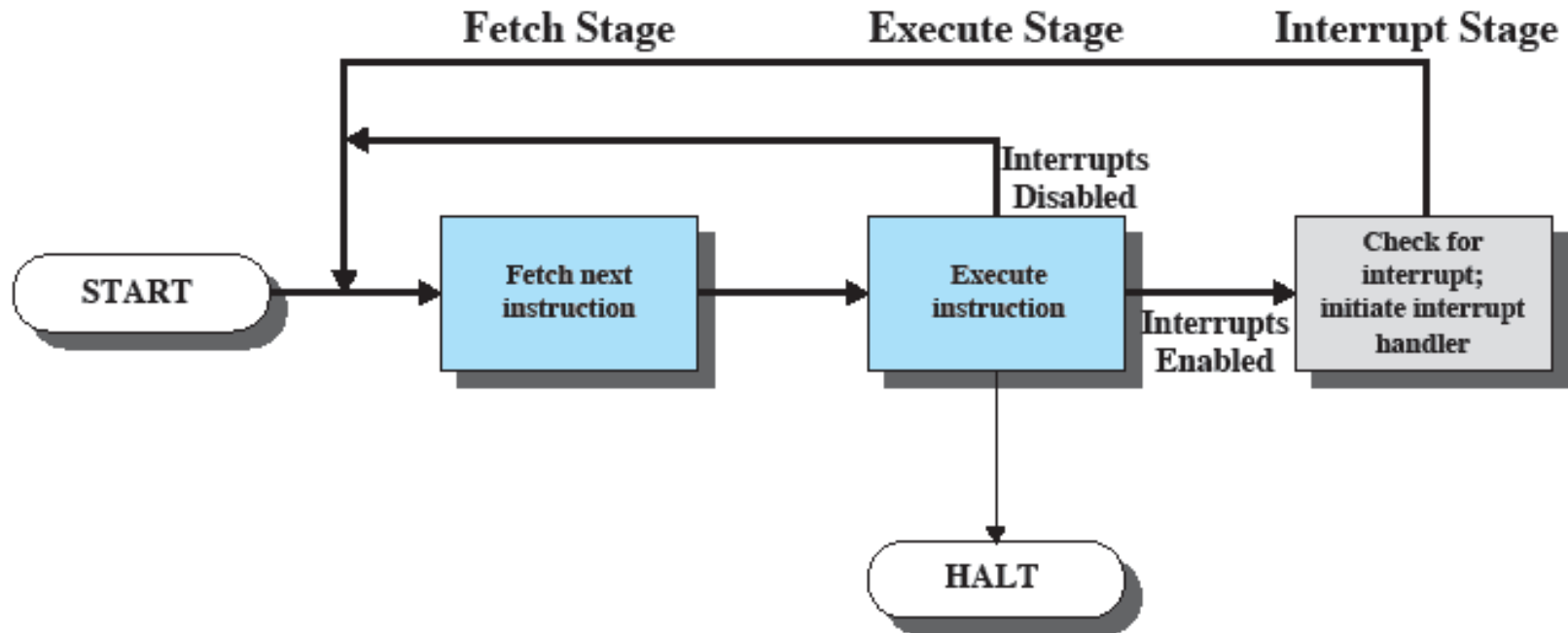


Figure 1.7 Instruction Cycle with Interrupts

Proračun vremena izvršavanja programa: kratko čekanje U/I

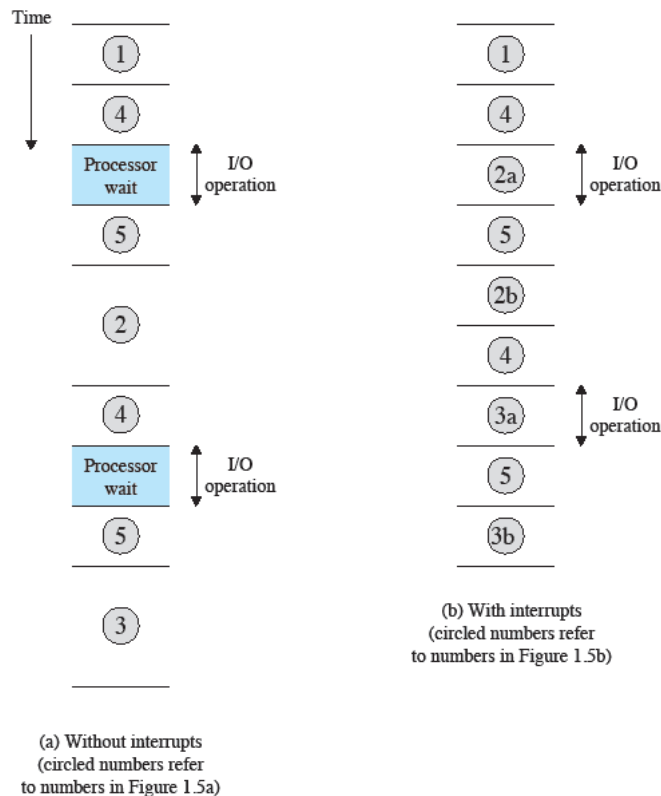


Figure 1.8 Program Timing: Short I/O Wait

Proračun vremena izvršavanja programa: dugo čekanje U/I

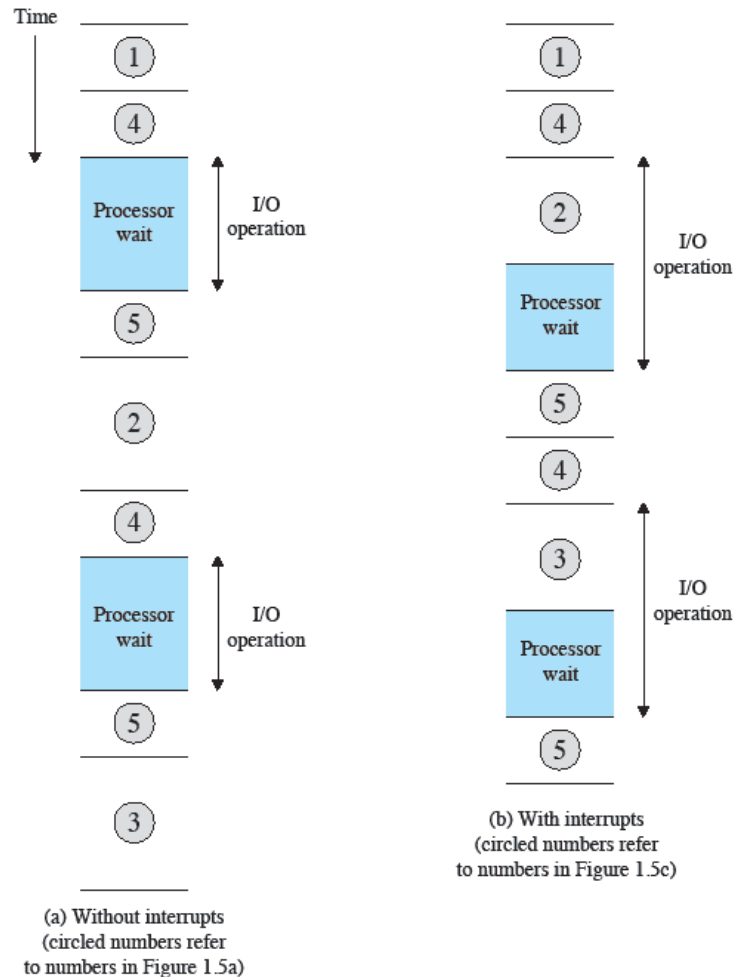
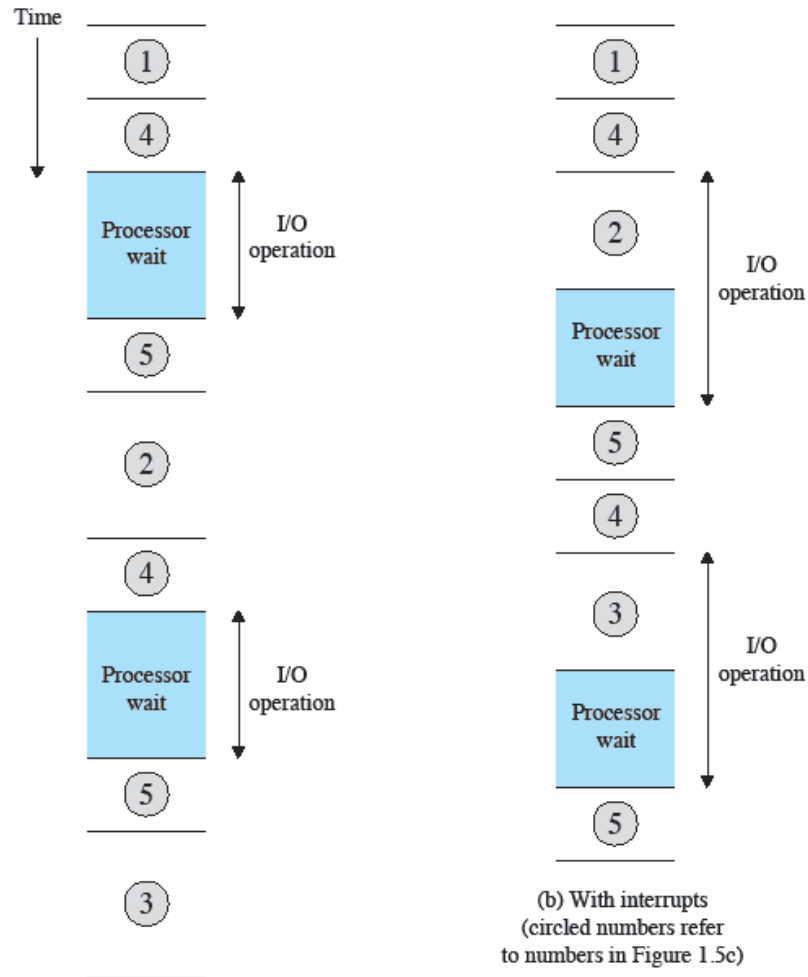


Figure 1.9 Program Timing: Long I/O Wait

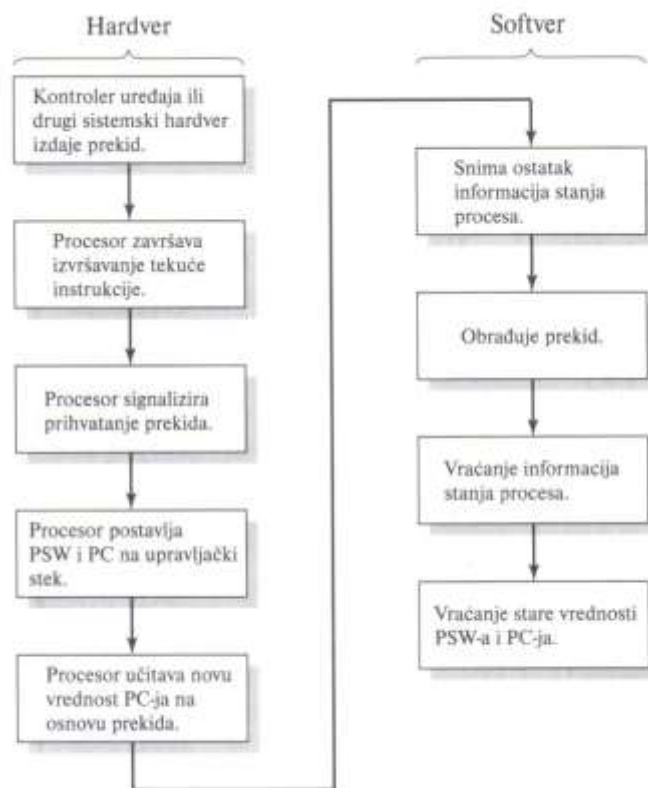
Program Timing: Long I/O Wait



(a) Without interrupts
(circled numbers refer to numbers in Figure 1.5a)

Figure 1.9 Program Timing: Long I/O Wait

Jednostavna obrada prekida



Slika 1.10 Jednostavna obrada prekida.

Simple Interrupt Processing

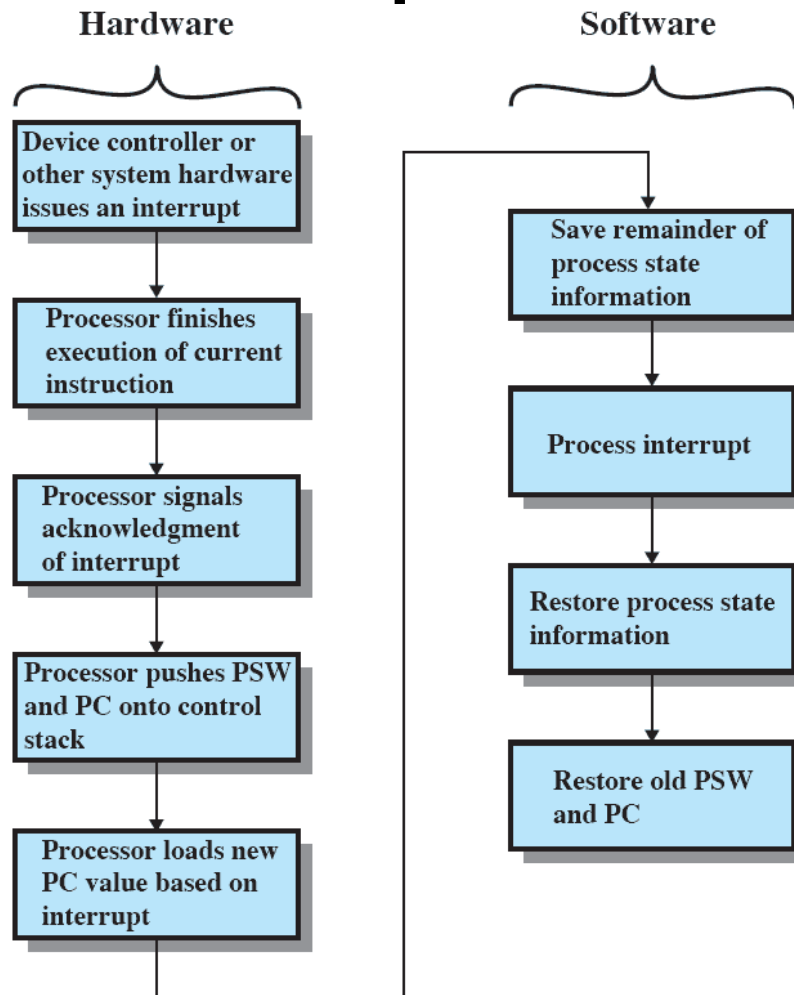
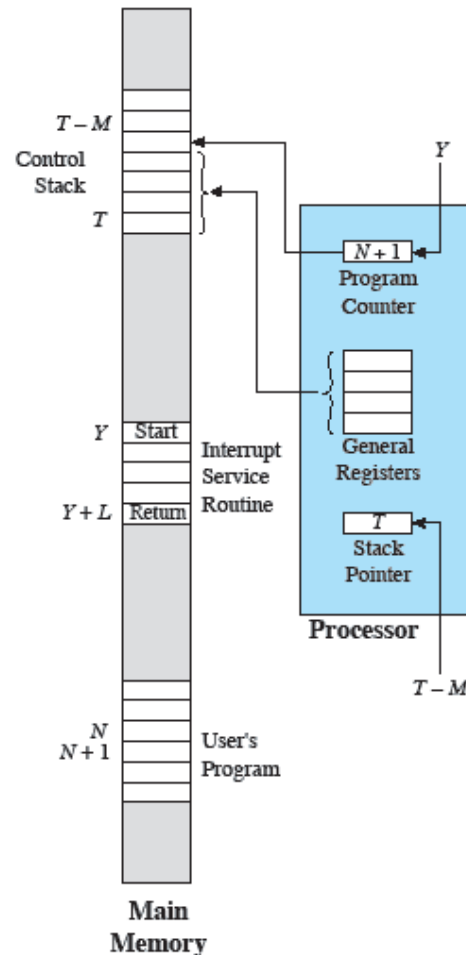


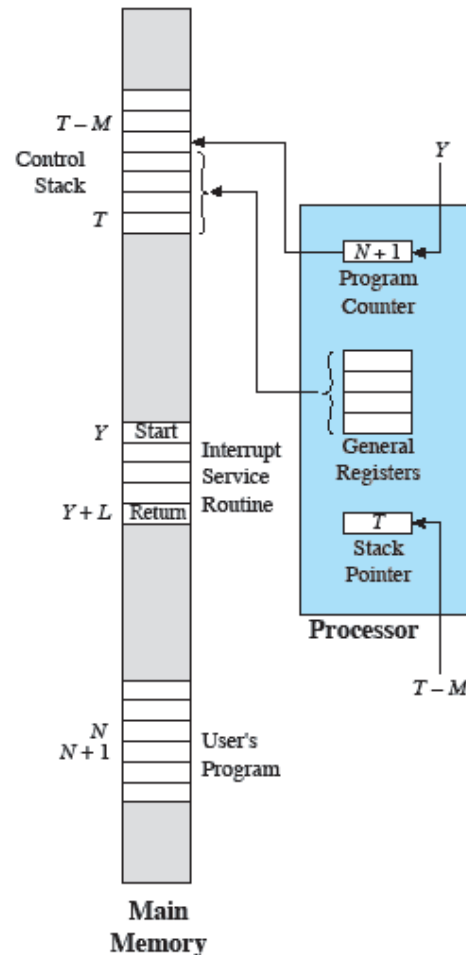
Figure 1.10 Simple Interrupt Processing

Promene u memoriji i registrima pri prekidu



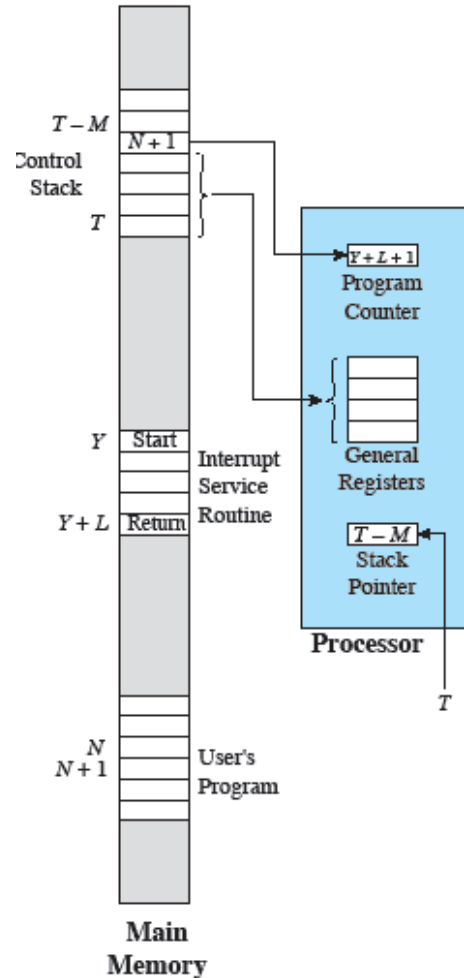
(a) Interrupt occurs after instruction at location N

Changes in Memory and Registers for an Interrupt



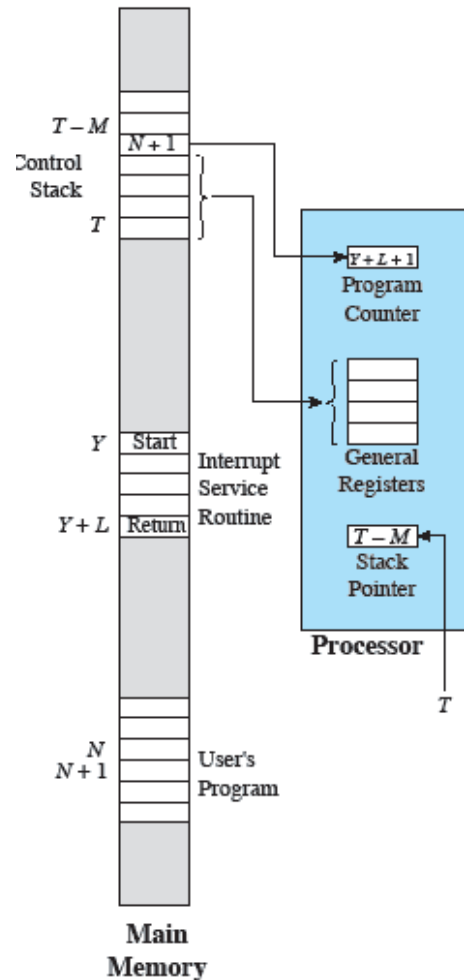
(a) Interrupt occurs after instruction at location N

Changes in Memory and Registers for an Interrupt

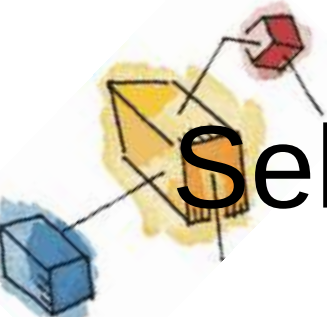


(b) Return from interrupt

Promene u memoriji i registrima pri prekidu

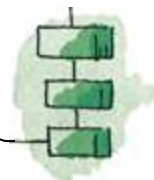


(b) Return from interrupt

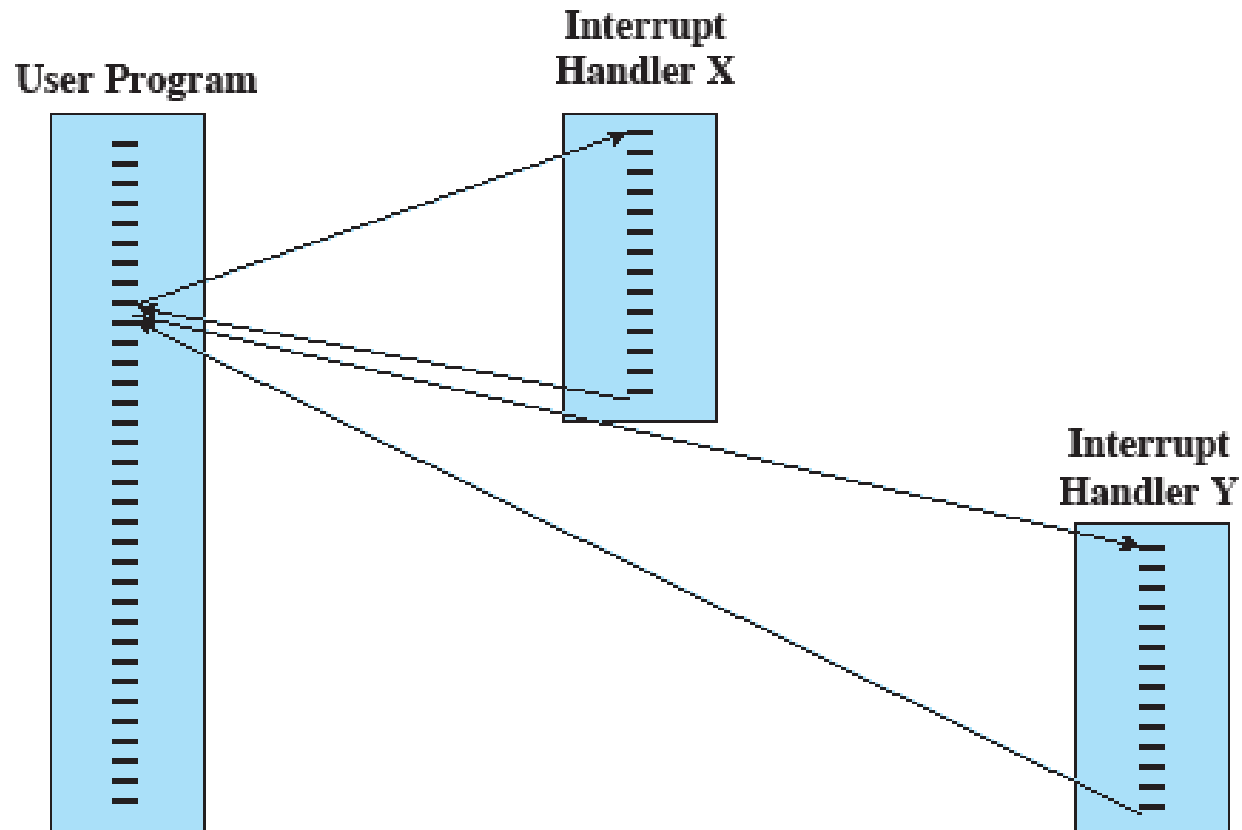


Sekvencijalna obrada prekida

- Prvi pristup:
 - onemogućavanje prekida u vreme obrade prekida
- Prekidi se obrađuju strogo prema redosledu;
- Loša strana:
 - ne vodi se računa o relativnom prioritetu ili
 - vremenski kritičnim potrebama.

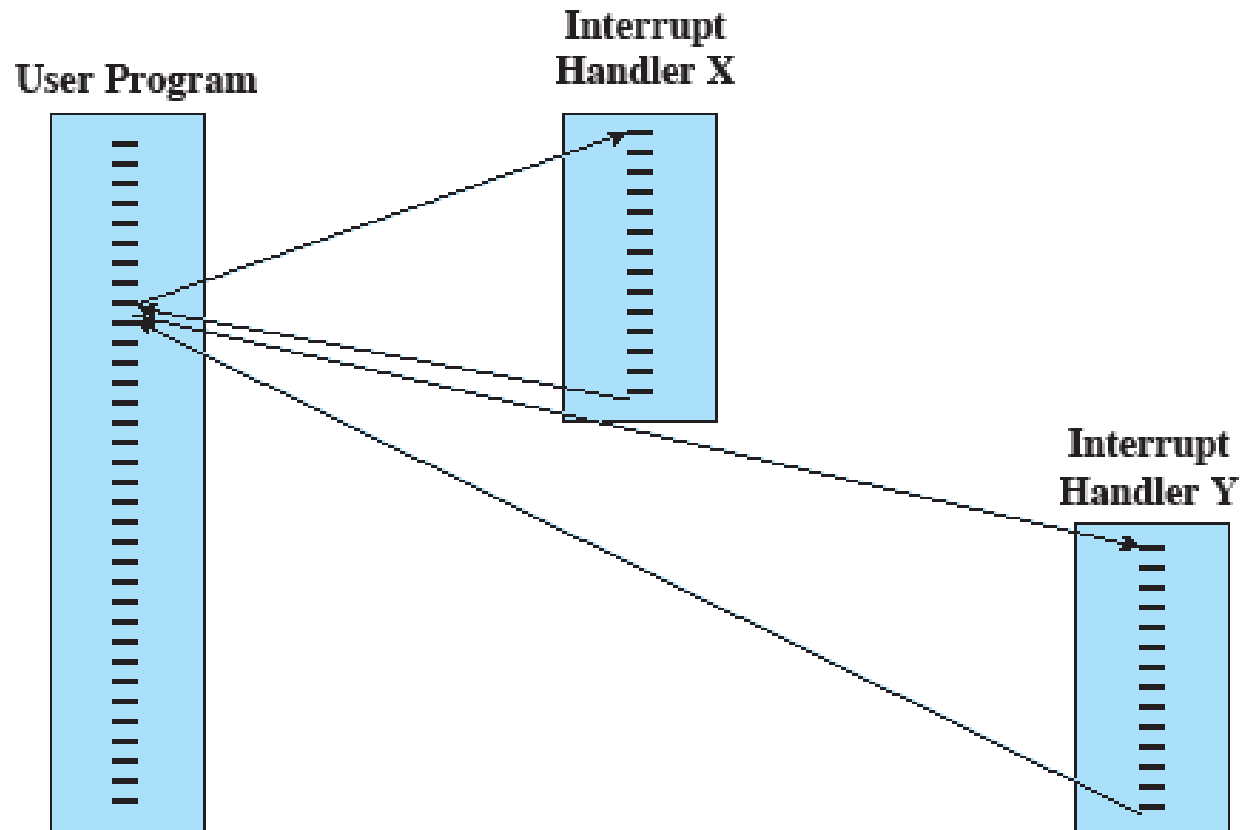


Sekvencijalna obrada prekida



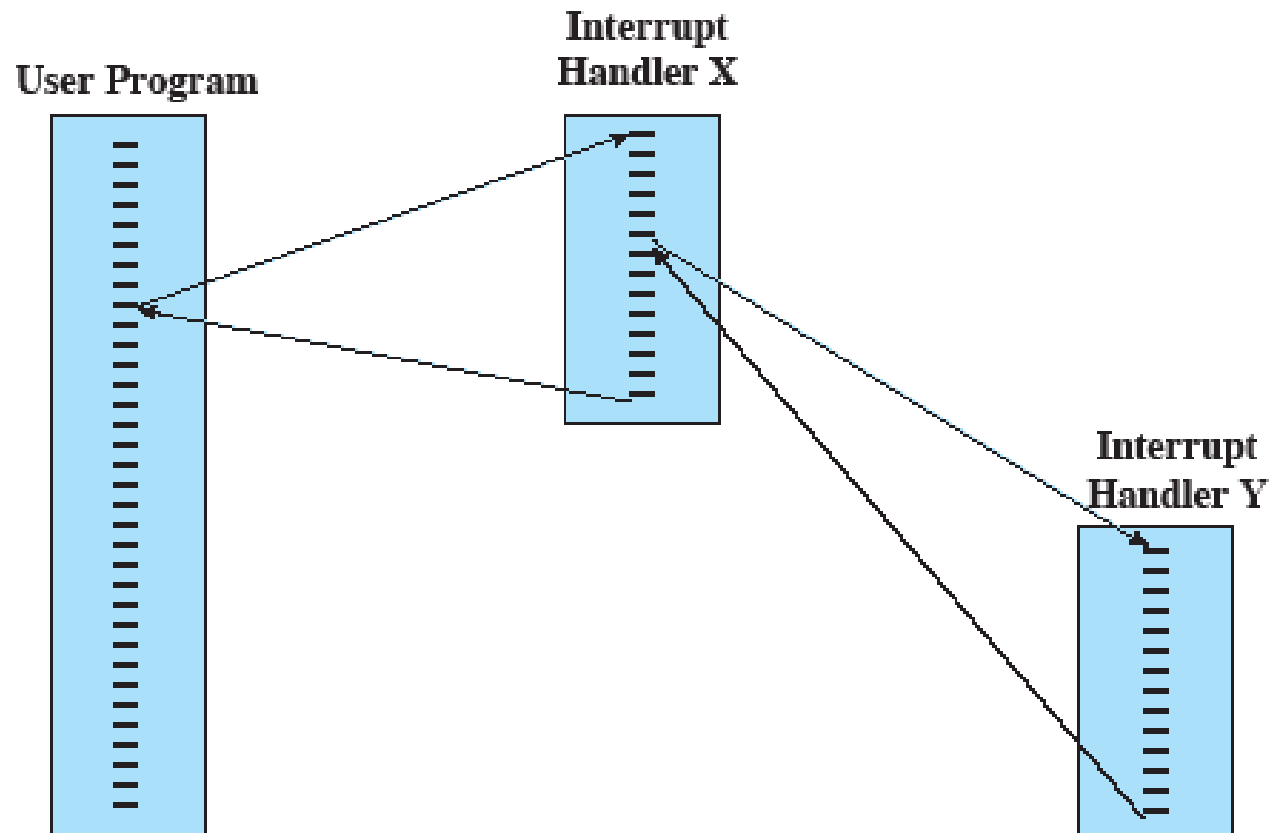
(a) Sequential interrupt processing

Sequential Interrupt Processing

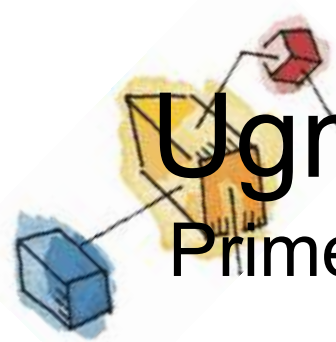


(a) Sequential interrupt processing

Ugnježdjena obrada prekida



(b) Nested interrupt processing



Ugnježdjena obrada prekida (2)

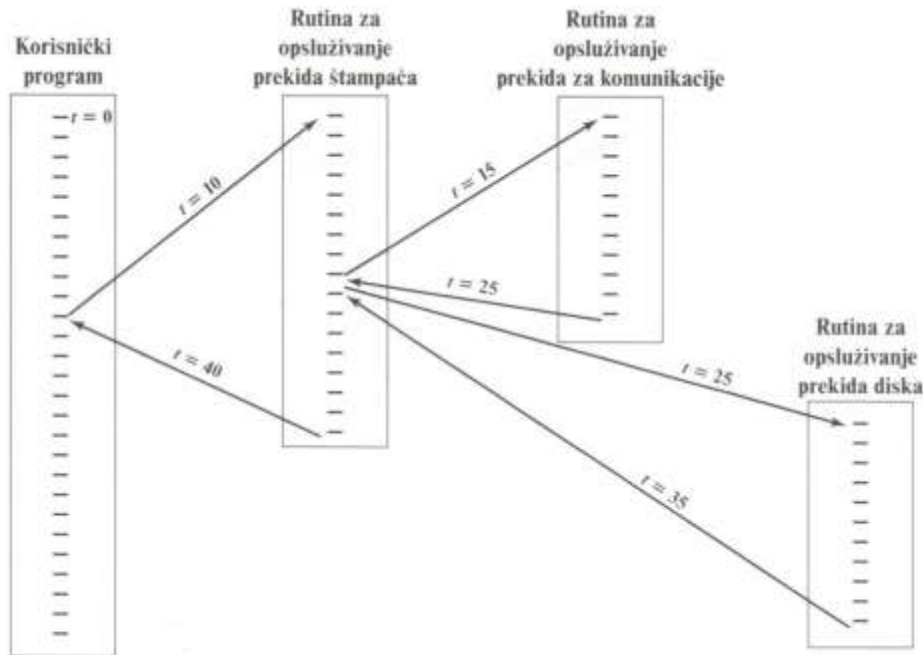
Primer vremenske sekvence višestrukih prekida

- Drugi pristup
 - da se definišu prioriteti prekida i
 - da se dozvoli da prioritet sa višim prioritetom izazove prekid prekida sa nižim prioritetom.

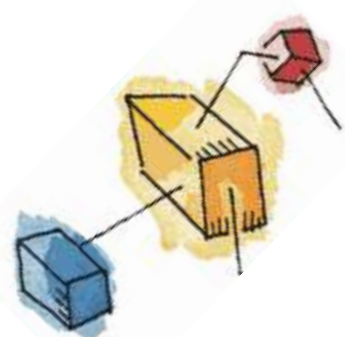


Ugnježdjena obrada prekida (3)

Primer vremenske sekvence višestrukih prekida

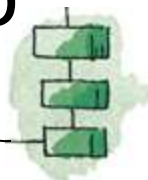


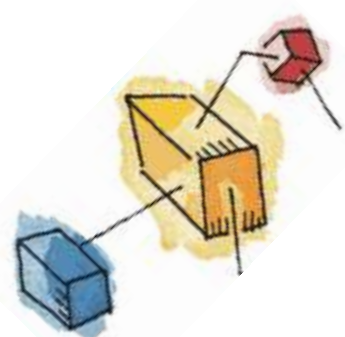
U/I uređaj	Nivo priori	Vreme pojavlj t=
Štampač	2	10
Disk	4	20
Komunikac. uređaj	5	15



Multiprogramiranje

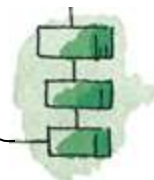
- Procesor ima dva ili više programa za izvršavanje
- Redosled izvršavanja programa zavisiće od njihovog relativnog prioriteta kao i od toga da li čekaju na U/I
- Po završetku rutine za obradu prekida upravljanje možda neće biti odmah vraćeno korisničkom programu koji se u to vreme izvršavao

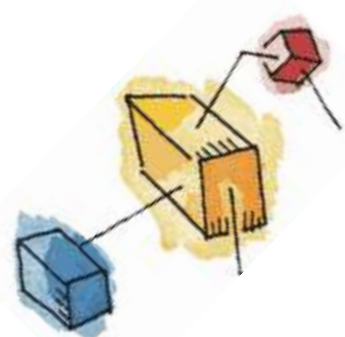




Multiprogramming

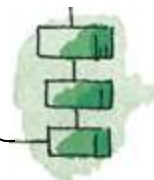
- Processor has more than one program to execute
- The sequence in which programs are executed depend on their relative priority and whether they are waiting for I/O
- After an interrupt handler completes, control may not return to the program that was executing at the time of the interrupt

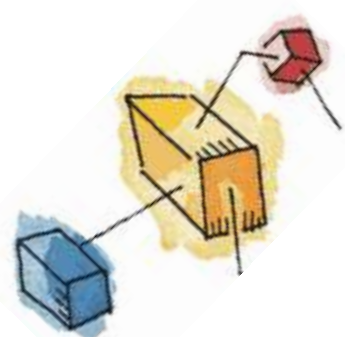




1.5 Hijerarhija memorije

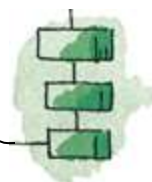
- Ograničenja pri projektovanju:
 1. koliko memorije;
 2. koliko da bude brza i
 3. koliko da košta?

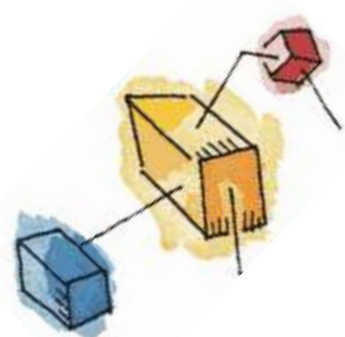




1.5 Hijerarhija memorije

- Postojanje KOMPROMISA između tri ključne karakteristike:
 1. cena;
 2. kapacitet i
 3. vreme pristupa.

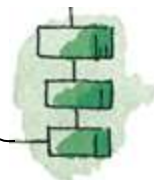


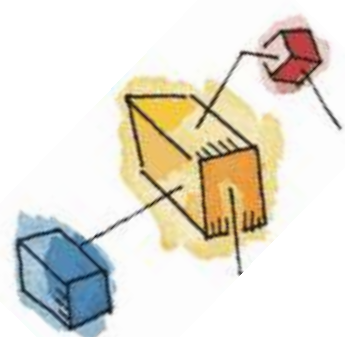


1.5 Hijerarhija memorije

Za širok asortiman tehnologija važe sledeći odnosi:

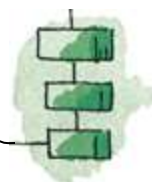
- Brže vreme pristupa, veća cena po bitu
- Veći kapacitet, manji troškovi po bitu
- Veći kapacitet, manja brzina pristupa





Memory Hierarchy

- Faster access time, greater cost per bit
- Greater capacity, smaller cost per bit
- Greater capacity, slower access speed



Hijerarhija memorije

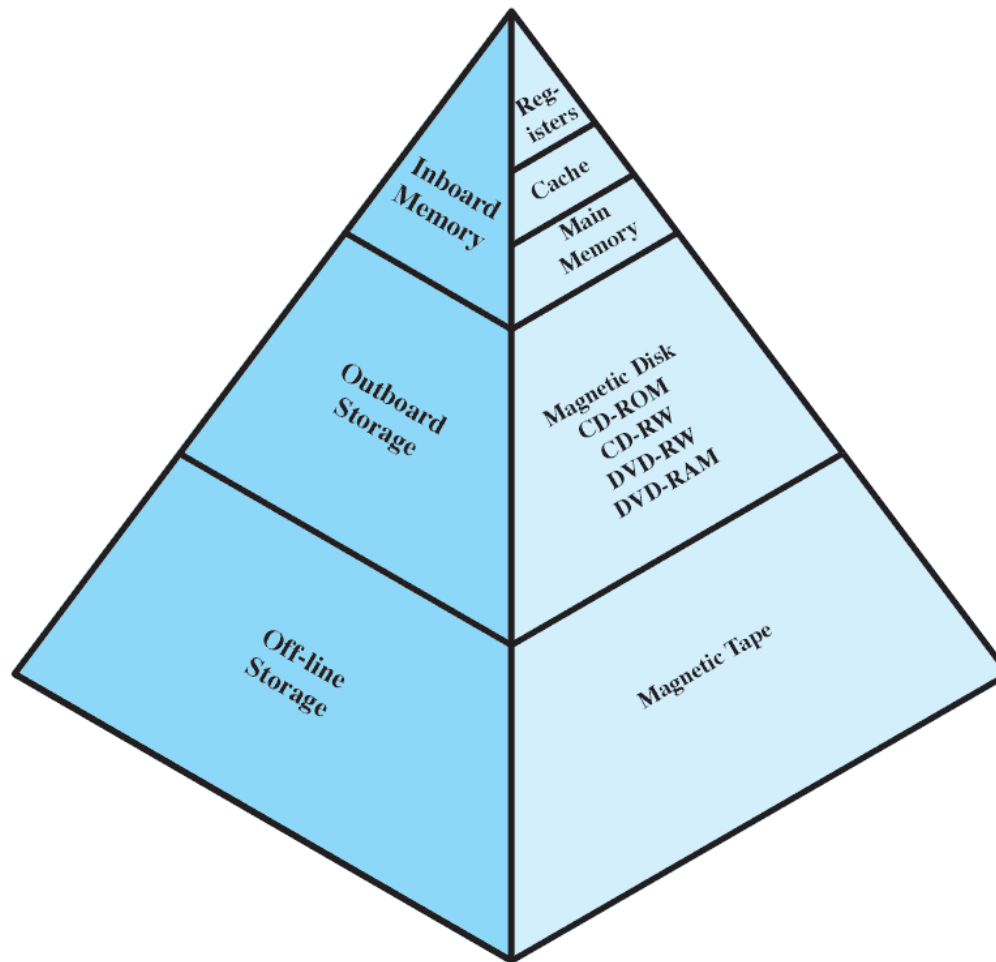


Figure 1.14 The Memory Hierarchy

The Memory Hierarchy

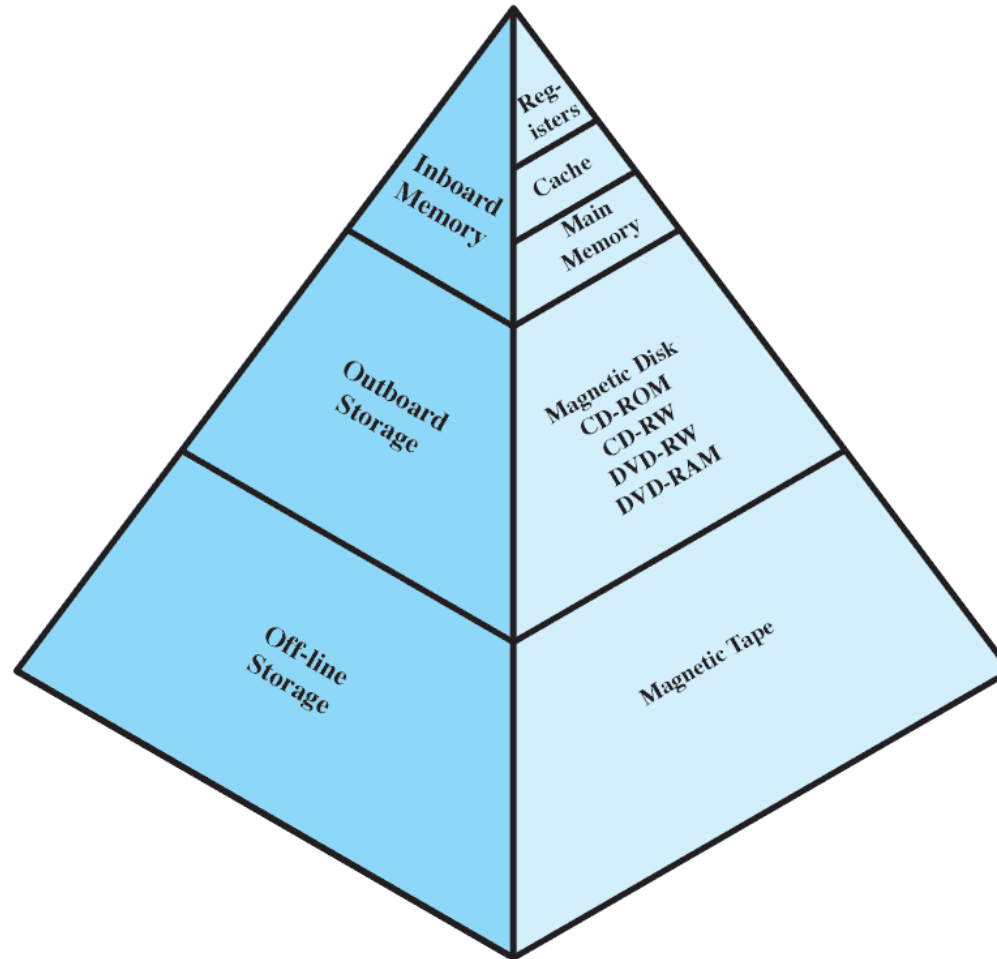
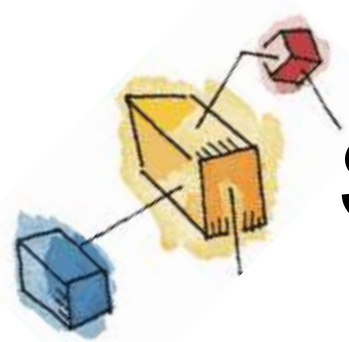


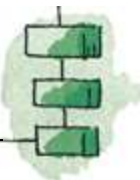
Figure 1.14 The Memory Hierarchy

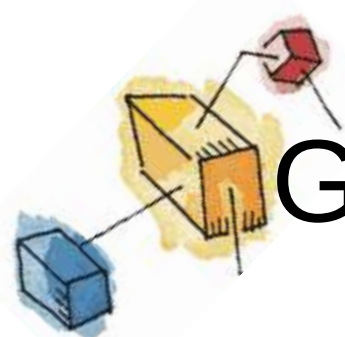


Spuštanje niz hijerarhiju

- Smanjuje se cena po bitu
- Povećava se kapacitet
- Povećava se vreme pristupa
- Smanjuje se učestanost pristupa procesora memoriji

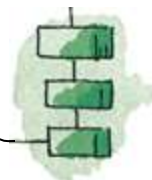
- Str.27

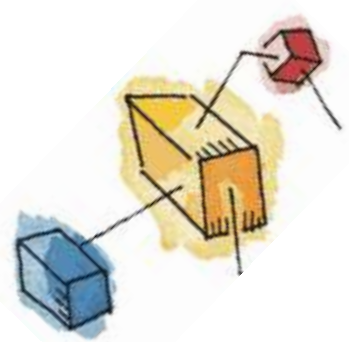




Going Down the Hierarchy

- Decreasing cost per bit
- Increasing capacity
- Increasing access time
- Decreasing frequency of access to the memory by the processor

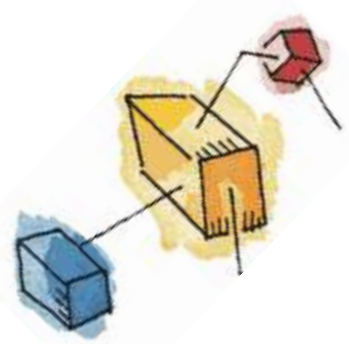




Sekundarna memorija

- Pomoćna memorija
- Spoljna (*External*)
- Trajna (*Nonvolatile*)
- Koristi se za skladištenje programa i datoteka sa podacima
 - **Virtuelna memorija**, kao proširenje glavne memorije

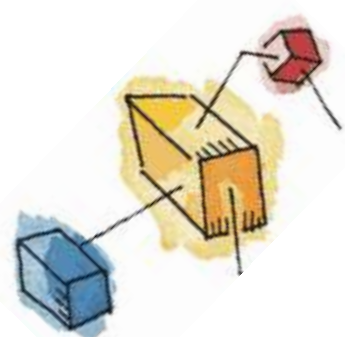




Secondary Memory

- Auxiliary memory
- External
- Nonvolatile
- Used to store program and data files

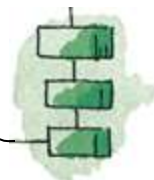




1.6 Keš memorija

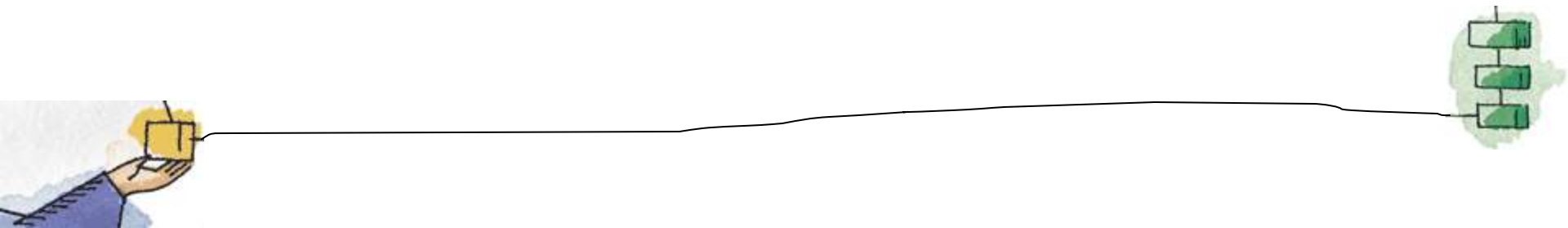
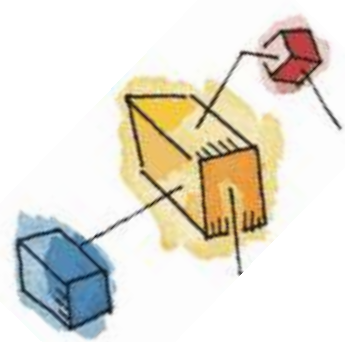
Motivacija:

- Brzina procesora je veća od brzine pristupa memoriji
- Koristi (*Exploit*) princip lokalnosti obezbeđivanjem male, brze memorije
 - Smeštene između registara procesora i glavne mem
 - sa ciljem poboljšanja performansi.



Cache Memory

- Processor speed faster than memory access speed
- Exploit the principle of locality with a small fast memory



Keš i glavna memorija

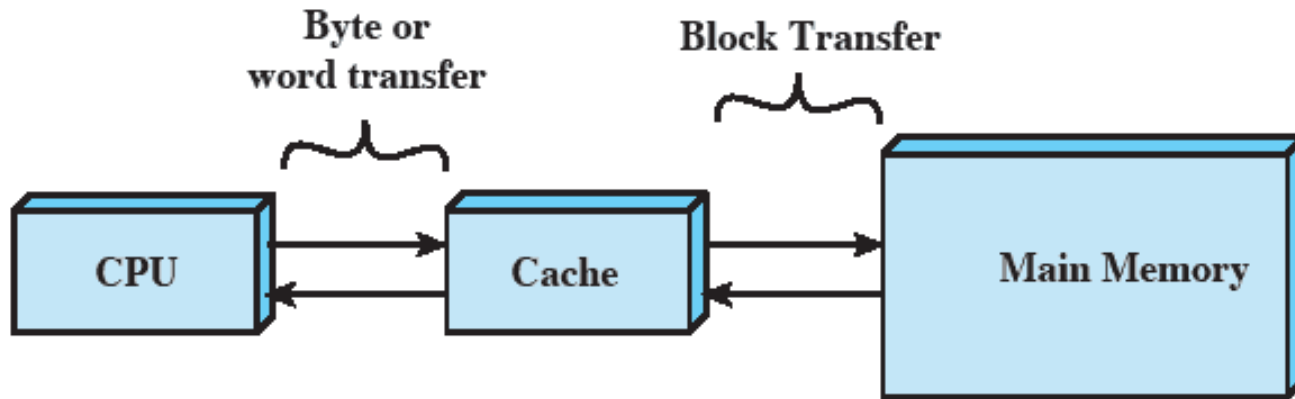


Figure 1.16 Cache and Main Memory

Cache and Main Memory

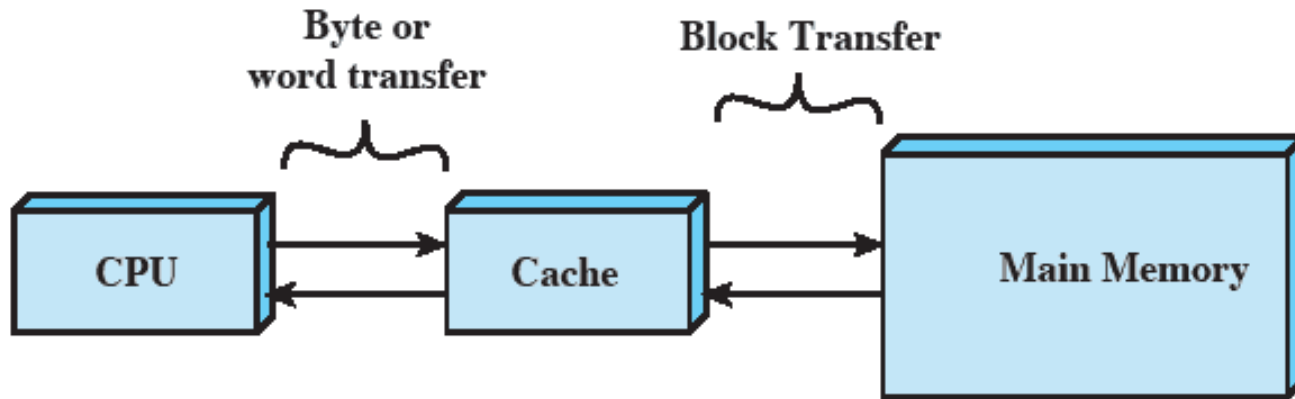
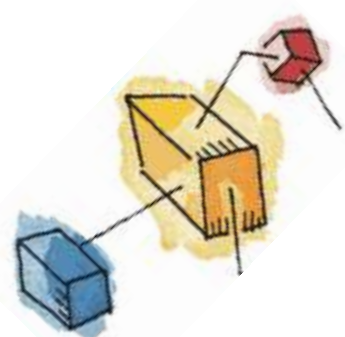


Figure 1.16 Cache and Main Memory



Principli keš memorije

- Sadrži kopiju dela glavne memorije
- Procesor najpre proverava keš
- Ako željeni podatak nije nađen, relevantni blok memorije učitava se u keš
- Zbog fenomena lokalnosti adrese, verovatno je da će mnoge buduće adrese biti u bloku
- Str.31

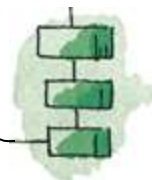
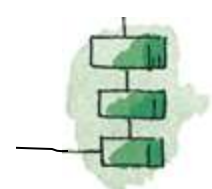
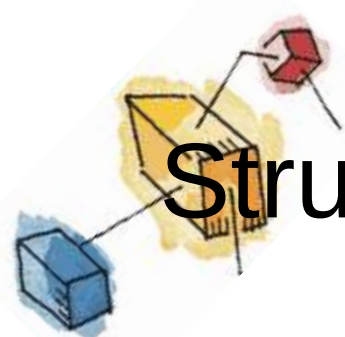




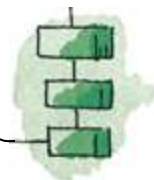
Figure 1.17 Cache/Main-Memory Structure

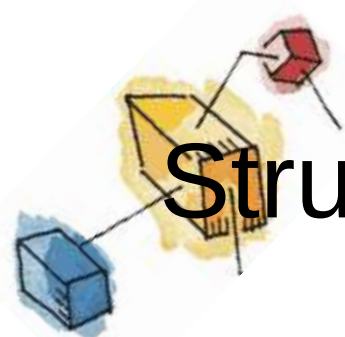




Struktura keš/glavna memorija (2)

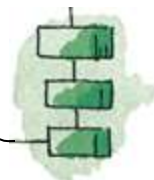
- Pojašnjenje:
 - glavna memorija = 2^n adresibilnih reči;
 - svaka reč ima jedinstvenu n -bitnu adresu;
 - blokovi su fiksne dužine od K reči,
 - broj blokova $M=2^n/K$;
 - keš memorija se sasoji od
 - C slotova koji imaju
 - K reči ($C \ll M$);
 - tag- ukazuje koji je trenutno blok smešten u nju.





Struktura keš/glavna memorija (3)

- Pojašnjenje:
 - tag- viši deo bitova adrese:
 - Primer: 6-bitna adresa i 2-bitni tag
 - tag 01 odnosi se na:
 - 010000,
 - 010001,
 - 010010,
 - 010011,
 -
 - 011111.



Operacija čitanja keš memorije

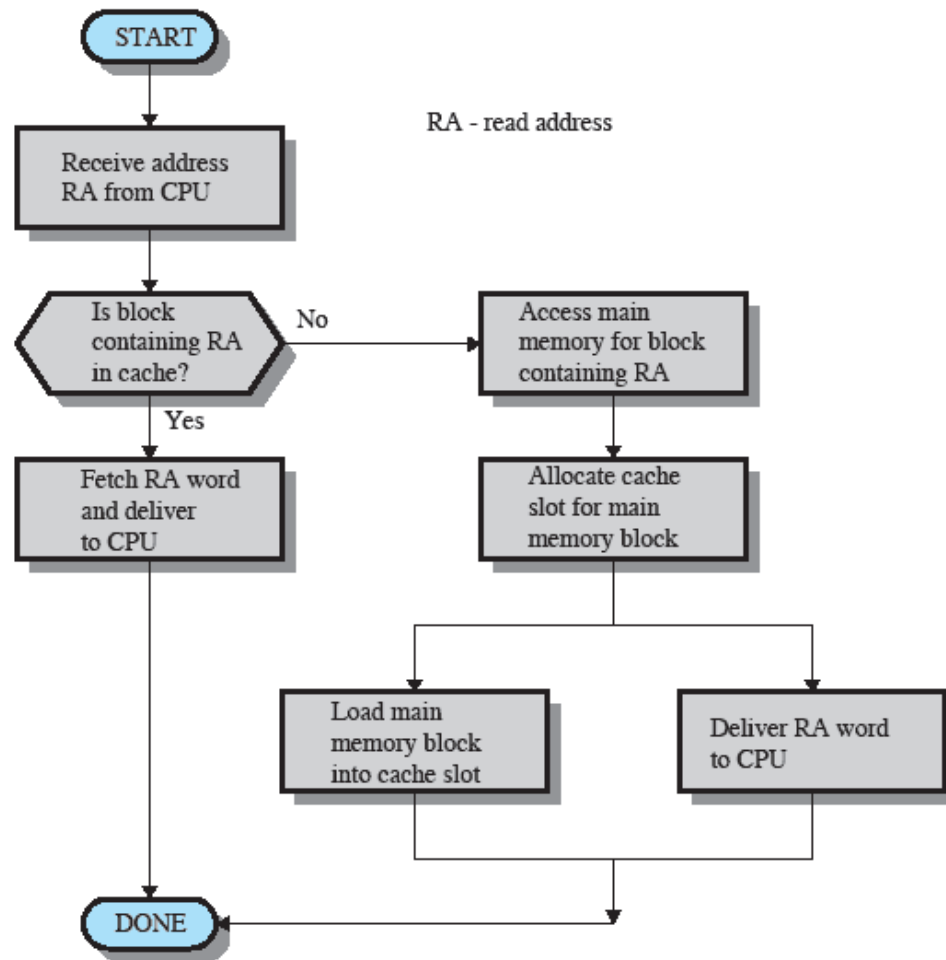
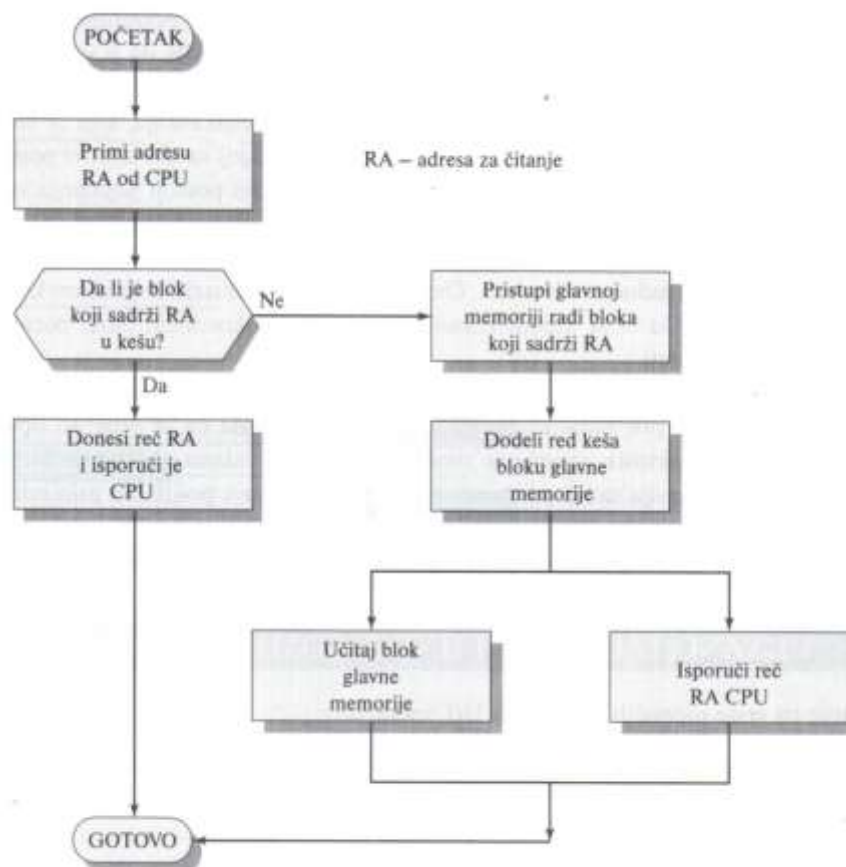
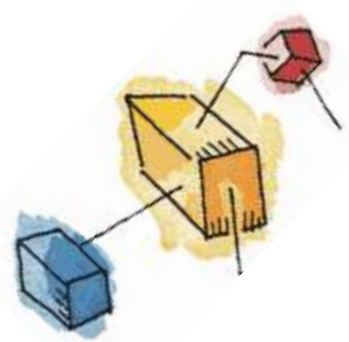


Figure 1.18 Cache Read Operation

Operacija čitanja keš memorije

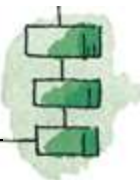


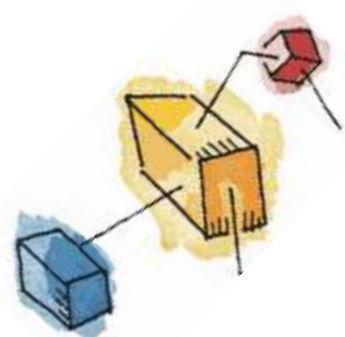
Slika 1.18 Operacija čitanja keš memorije



Dizajn keš memorije

- Problemi dizajna keš memorije:
 1. Veličina keš memorije
 2. Veličina bloka
 3. Funkcija preslikavanja
 4. Algoritam za zamenu
 5. Politika upisa

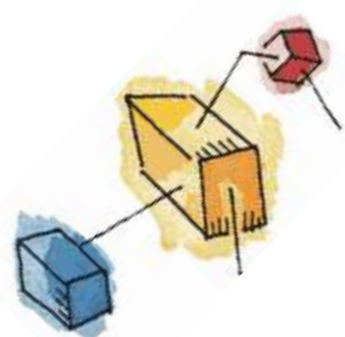




Dizajn keš memorije

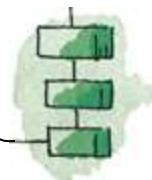
- Veličina keš memorije
 - I mala keš memorija može imati značajan uticaj na performanse
- Veličina bloka
 - Jedinica podataka koja se može razmeniti između keša i glavne memorije
 - Povećanje veličine bloka smanjuje verovatnoću pogotka dok verovatnoća korišćenja novih pribavljenih podataka postaje manja od verovatnoće ponovnog korišćenja podataka koji se moraju premestiti iz keša

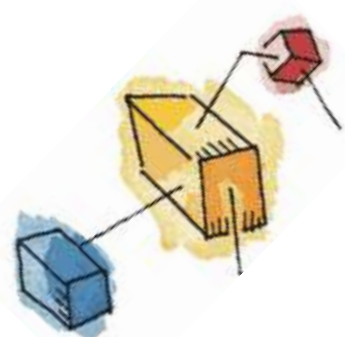




Cache Principles

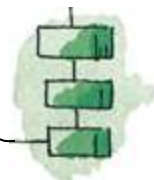
- Veličina keš memorije
 - Even small caches have significant impact on performance
- Block size
 - The unit of data exchanged between cache and main memory
 - Larger block size yields more hits until probability of using newly fetched data becomes less than the probability of reusing data that have to be moved out of cache





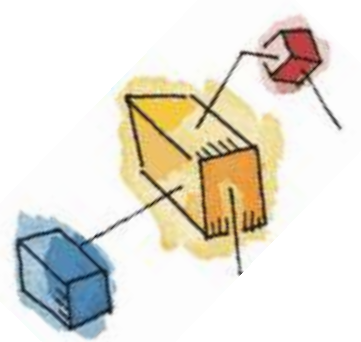
Dizajn keš memorije

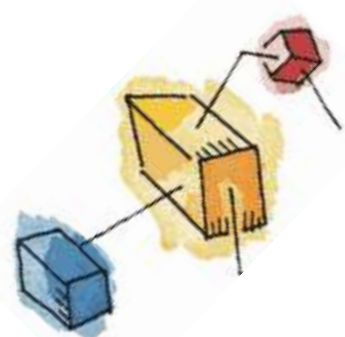
- Funkcija preslikavanja (*Mapping function*)
 - Utvrđuje koju će lokaciju u keš memoriji zauzeti blok
- Algoritam za zamenu
 - Bira koji će blok biti zamenjen
 - Najmanji skoro korišćeni (LRU- *Least-recently-used*) algoritam



Cache Principles

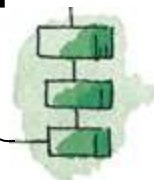
- Mapping function
 - Determines which cache location the block will occupy
- Replacement algorithm
 - Chooses which block to replace
 - Least-recently-used (LRU) algorithm

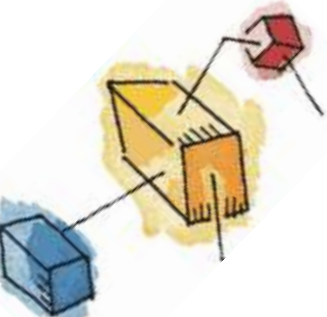




Dizajn keš memorije

- Politika upisa
 - Diktira kada će se odigrati operacija upisa u memoriju:
 1. Upis se može pojaviti pri svakom **ažuriranju bloka**
 2. Upis se javlja smo pri **zameni bloka**
 - **Minimizuju** se operacije upisa
 - Glavna morija se ostavlja u **zastarelom** stanju
 1. može remetiti višeprocorske operacije i
 2. direktan pristup memoriji preko hardverskih U/I modula

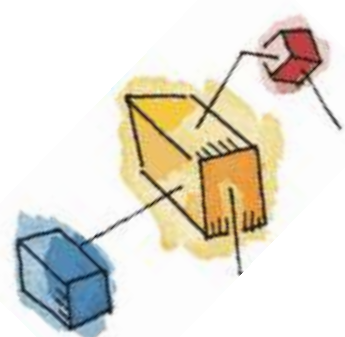




Cache Principles

- Write policy
 - Dictates when the memory write operation takes place
 - Can occur every time the block is updated
 - Can occur when the block is replaced
 - Minimize write operations
 - Leave main memory in an obsolete state

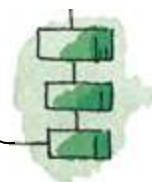


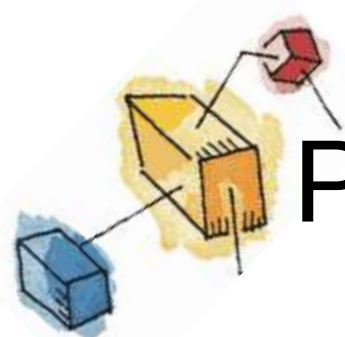


Ulazno/Izlazne tehnike komuniciranja

- Postoje TRI vrste mogućih tehnika za U/I operacije:
 1. Programirani U/I;
 2. U/I upravljani prekidima;
 3. Direktan pristup memoriji.

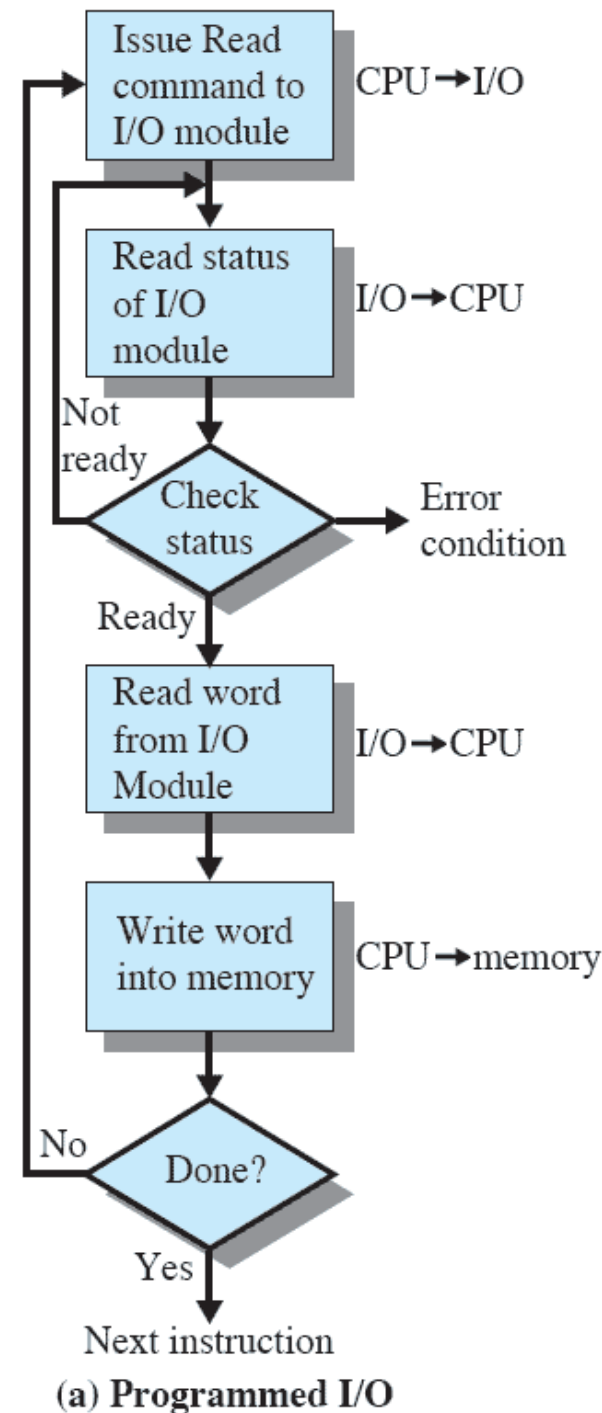
- Str.34

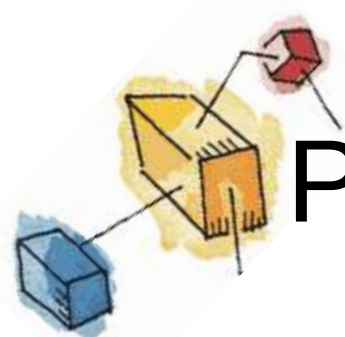




Programirani U/I

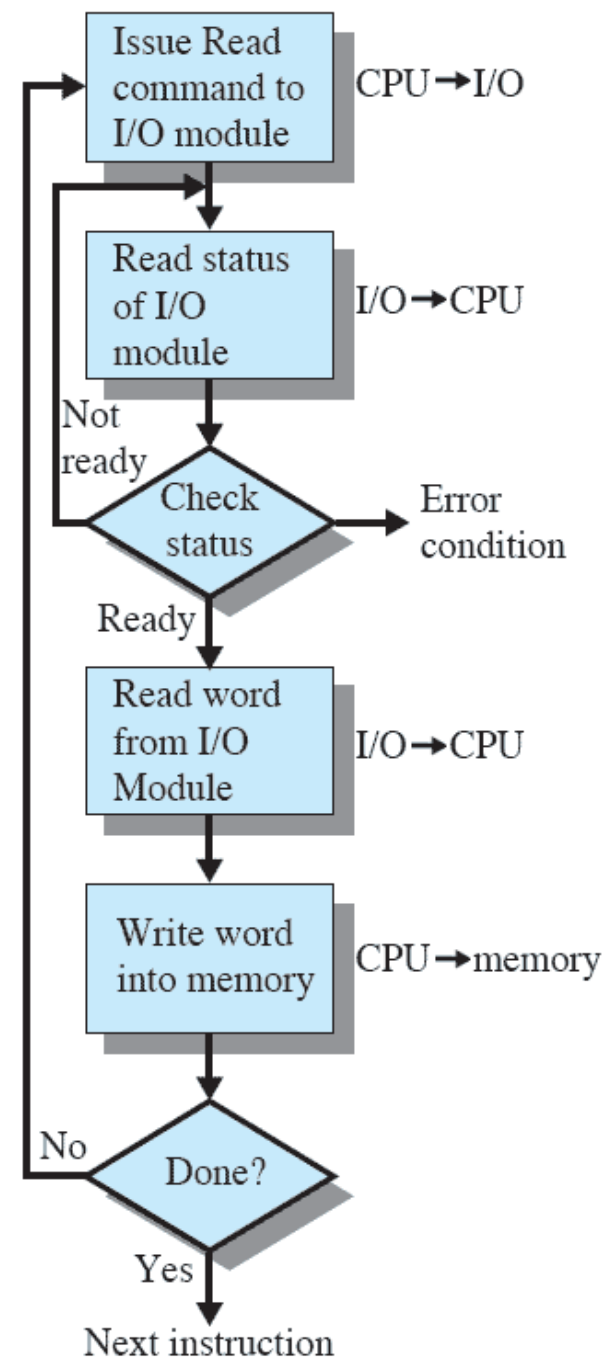
- U/I modul izvršava zahtevanu akciju **a ne procesor**
- Postavlja odgovarajuće bitove u U/I statusnom registru
- Ne generiše prekid
- Procesor **povremeno** proverava status U/I modula dok operacija ne bude završena



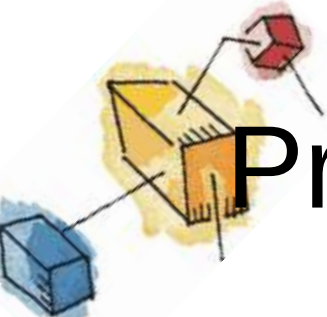


Programmed I/O

- I/O module performs the action, not the processor
- Sets the appropriate bits in the I/O status register
- No interrupts occur
- Processor checks status until operation is complete



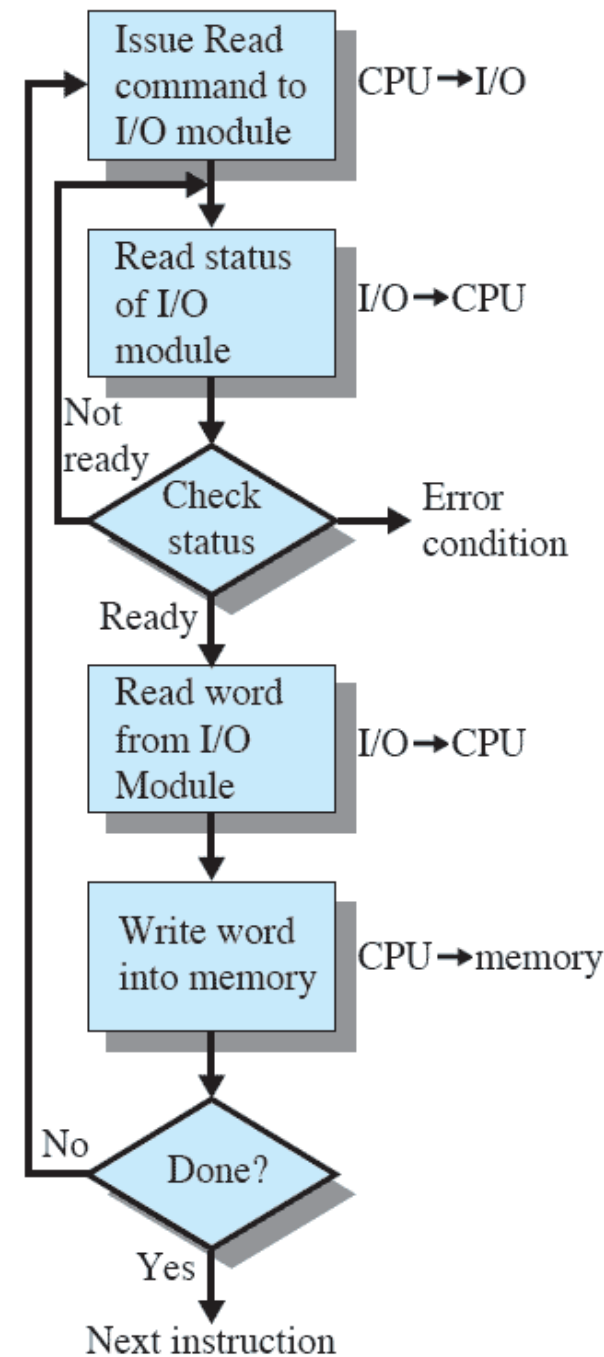
(a) Programmed I/O



Programirani U/I (2)

- Procesor je odgovoran za upis/čitanje podataka iz glavne memorije
- Skup potrebnih naredbi:
 1. Upravljačke-aktiviranje U/I uređaja;
 2. Statusne-testiranje uslova;
 3. Za prenos-registri procesora-eksterni uređaj.

- Str.34/35

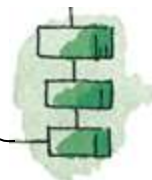
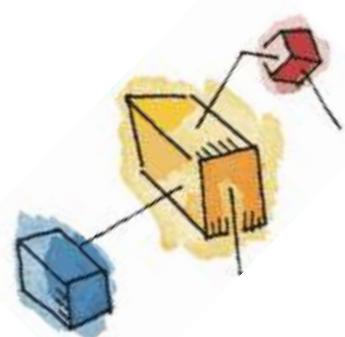


(a) Programmed I/O

Problemi

Programiranog U/I

- Procesor mora dugo **da čeka** da U/I bude spreman za prijem/predaju podataka
- Drugo rešenje:
 1. Procesor izda U/I naredbu modulu;
 2. Nastavlja sa drugim korisnim radom
 3. U/I modul će prekinuti procesor kada je spreman za razmenu podataka
- Str.36

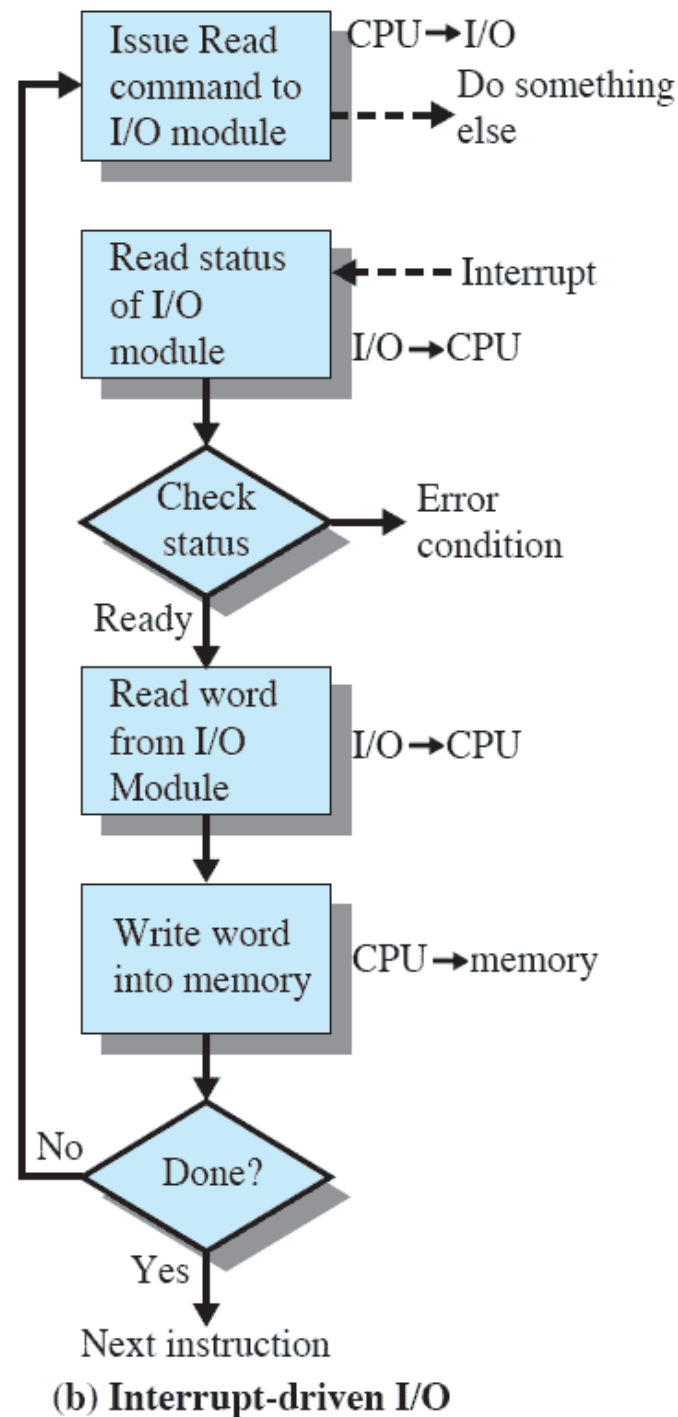


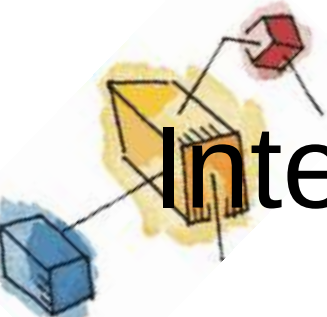


U/I upravljan prekidima

- Procesor je prekinut kada je U/I modul spreman za razmenu podataka
- Procesor snima sadržaj programa koji se izvršava i počinje izvršavanje rutine za obradu prekida (*interrupt-handler*)

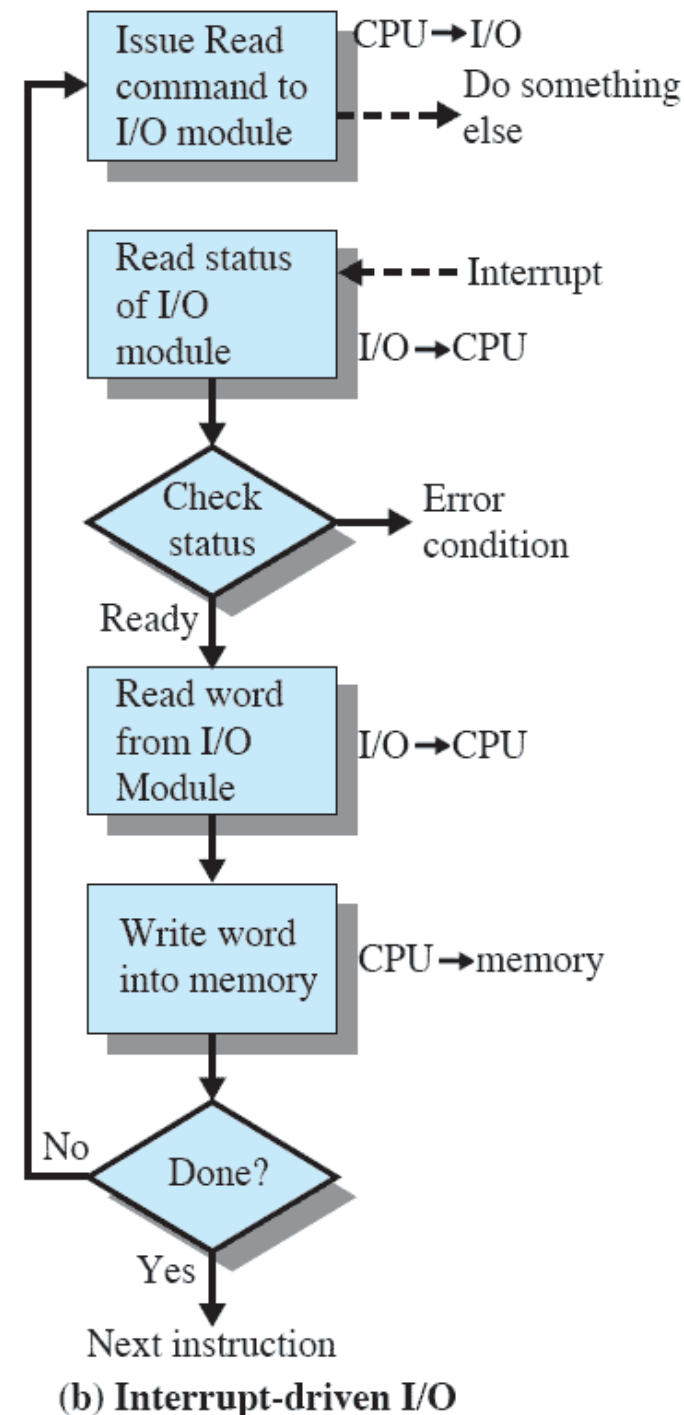
- Str.36





Interrupt-Driven I/O

- Processor is interrupted when I/O module ready to exchange data
- Processor saves context of program executing and begins executing interrupt-handler





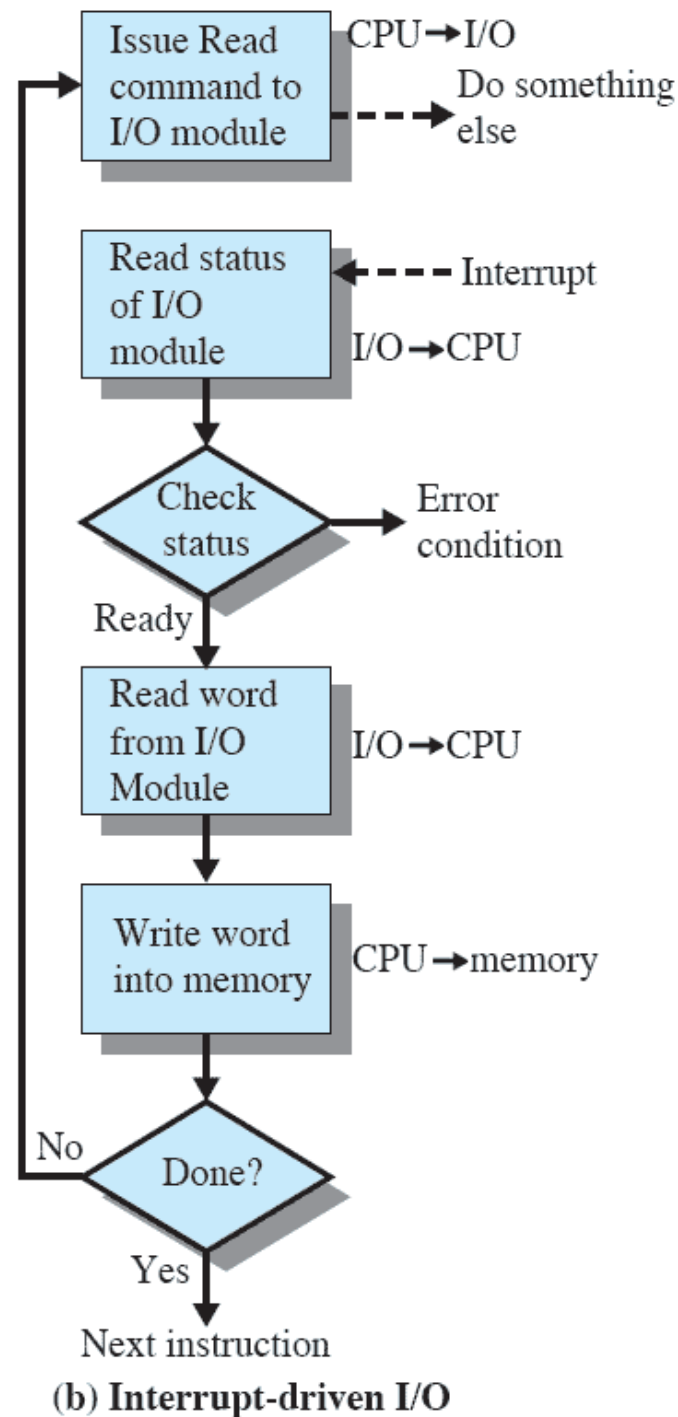
U/I upravljan prekidima

- Eliminiše bespotrebno čekanje

I dalje:

- Troši mnogo vremena jer svaka reč pročitana/zapisana prolazi kroz procesor

- Str.36



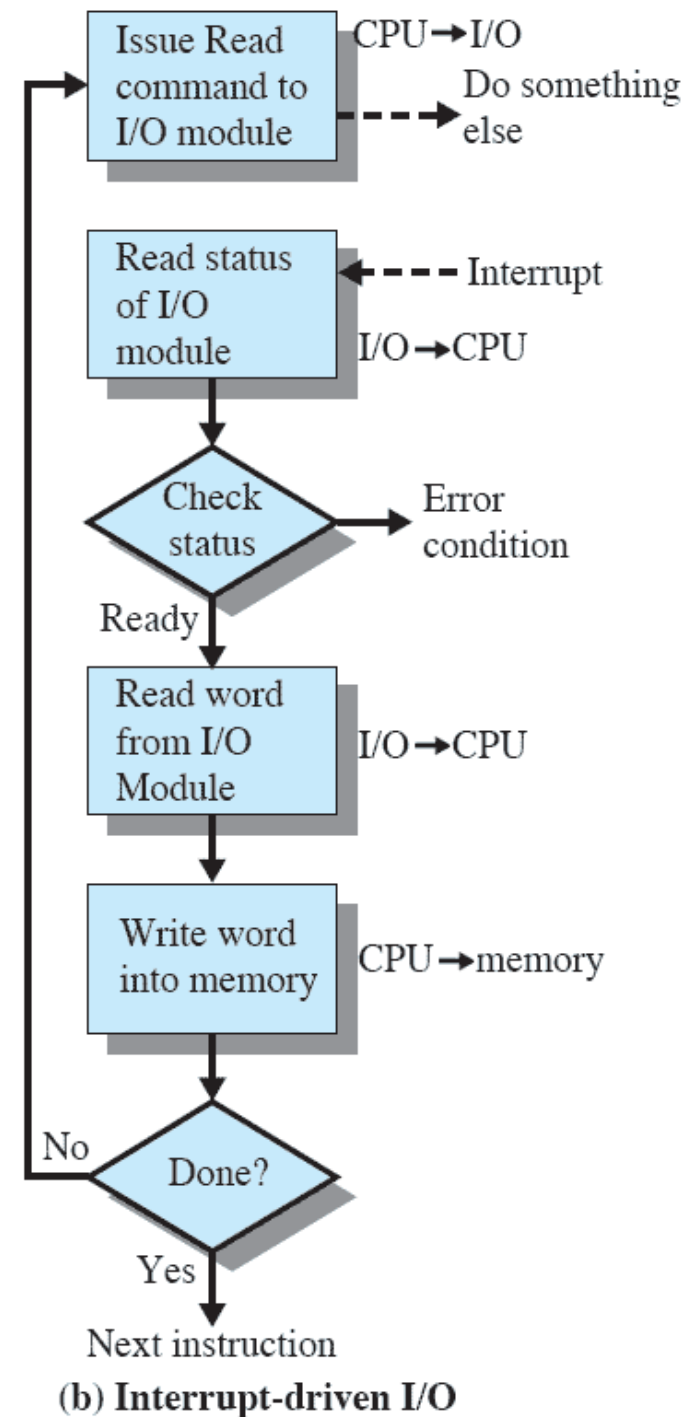


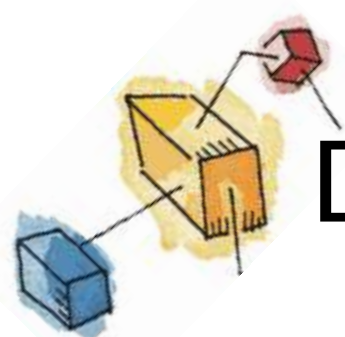
U/I upravljan prekidima

- Neophodno postojanje mehanizama koji će omogućiti procesoru

**da utvrdi poreklo
prekidnog signala**

- Str.36

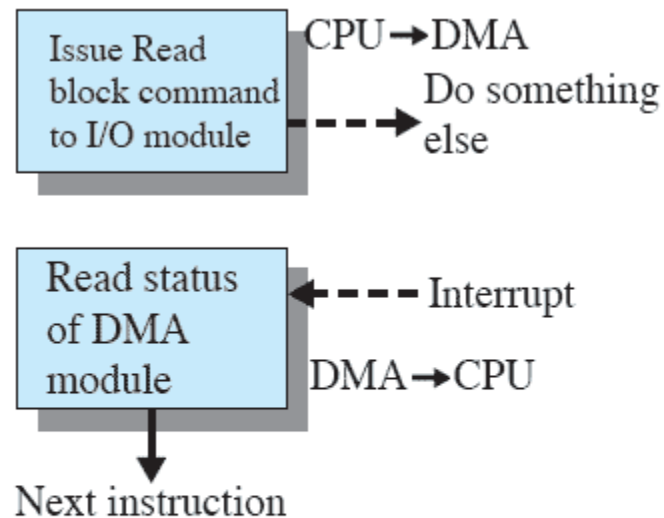




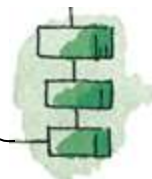
Direktan pristup memoriji

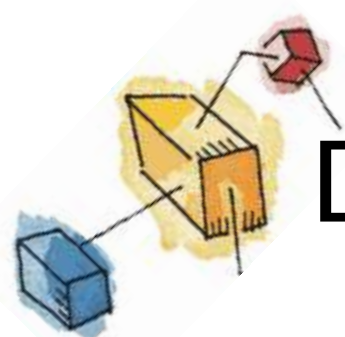
DMA modul:

- Prebacuje blok podataka **direktno** u memoriju ili iz nje
- Šalje prekidni signal procesoru po završetku prenosa
- Str.37



(c) Direct memory access





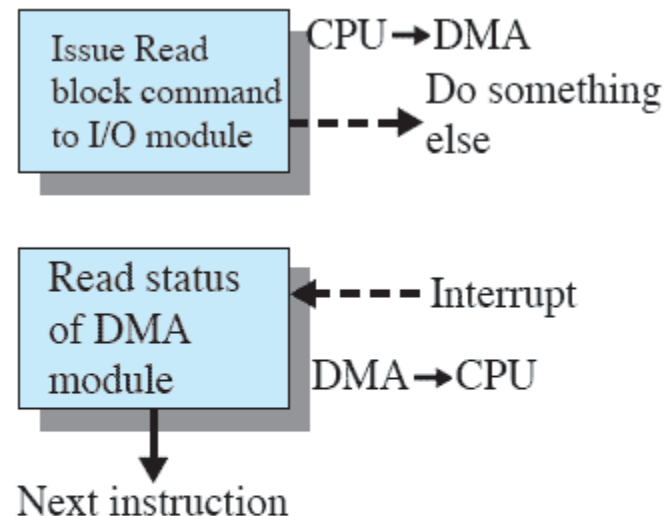
Direktan pristup memoriji

➔ procesor je uključen samo na početku i na kraju prenosa

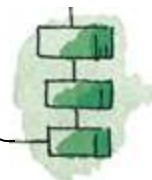
- Mnogo efikasniji prenos

Nedostatak:

1. Konkurencija za korišćenje magistrale
2. Procesor pravi pauzu od jednog ciklusa magistrale



(c) Direct memory access



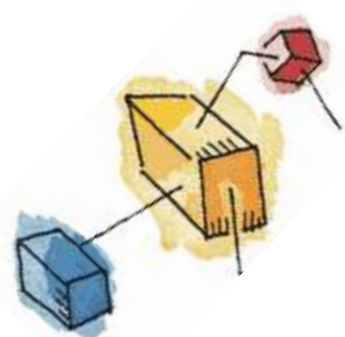


-
- ```

graph TD
 CPU1[CPU] -- "Issue Read block command to I/O module" --> IO[]
 IO -- "DMA" --> CPU2[CPU]
 CPU2 -- "Do something else" --> CPU2
 CPU2 -- "Interrupt" --> IO
 IO -- "DMA" --> CPU3[CPU]
 CPU3 -- "Read status of DMA module" --> CPU4[CPU]
 CPU4 -- "Next instruction" --> CPU4

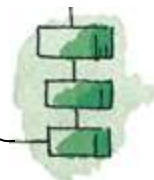
```
- The diagram illustrates the sequence of operations for a DMA transfer. It shows the interaction between the CPU and the I/O module (DMA) during a read operation. The process starts with the CPU issuing a read block command to the I/O module. The CPU then does something else while the DMA module handles the transfer. Once the transfer is complete, the DMA module sends an interrupt to the CPU. The CPU then reads the status of the DMA module and proceeds to the next instruction.

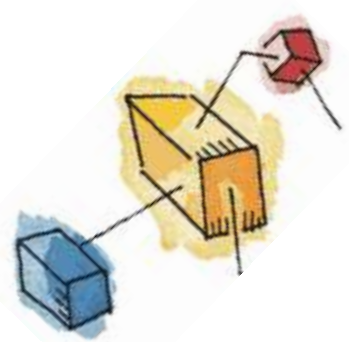




# Kontrolna pitanja

- 1.1 Nabrojte i kratko opišite četiri osnovna elementa računara.
- 1.2 Opišite dve glavne kategorije registara procesora.
- 1.3 Koje su četiri opšte različite akcije koje se mogu navesti mašinskim instrukcijama?
- 1.4 Šta je prekid?





# Kontrolna pitanja (2)

- 1.5 Kako se obrađuju višestruki prekidi?
- 1.6 Po kojim karakteristikama se razlikuju razni elementi u hijerarhiji memorije?
- 1.7 Šta je keš memorija?
- 1.8 Nabrojte i kratko opišite tri tehnike U/I operacija?

